CME 307: Optimization CME 307 / MS&E 311 / OIT 676

Lecture 10: Conic Optimization

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1 Overview

By now we know how to recognize a convex optimization problem, how to characterize its solutions, and how to build its Lagrange dual. These notes take up a practical question that sits between modeling and algorithms:

How should we hand a convex problem to a solver?

Our answer is a *structured representation*: we write the problem using affine maps together with a small number of standard convex cones. Along the way we meet

- *robust optimization*, which produces our first second-order cone constraint;
- the reasons solvers want conic structure, and when a different kind of solver is the better choice;
- *conic form* and the hierarchy LP $\subset$ QP $\subset$ SOCP $\subset$ SDP;
- the path from a model to a solver, including disciplined convex programming;
- *semidefinite programs* and *relaxations*, illustrated by experiment design and MaxCut;
- *conic duality*, which is one more application of the idea that a Lagrangian is built to be a lower bound.

Whenever we change the formulation of a problem, we will ask three questions:

what is preserved? what becomes easier to compute? what is certified?

Keep these questions in mind as you read. The final two sections collect the key concepts and a study guide for the quiz; the solutions to all exercises appear just before them.

2 Preliminaries: norms and symmetric matrices

This section collects the facts about norms and matrices that we use repeatedly. Most of it is review; read it quickly if it is familiar, and return to it when a later argument cites it.

2.1 Norms and the Cauchy–Schwarz inequality

For $x \in \mathbb{R}^n$ we use three norms:

$$\|x\|_1 = \sum_{i=1}^{n} |x_i|, \qquad \|x\|_2 = \Big(\sum_{i=1}^{n} x_i^2\Big)^{1/2}, \qquad \|x\|_\infty = \max_{i=1,\dots,n} |x_i|.$$

When we write $\|x\|$ without a subscript we mean the Euclidean norm $\|x\|_2$. Every norm is convex, nonnegative, and zero only at $x = 0$.

Proposition 2.1 (Cauchy–Schwarz)**.** *For all $u, v \in \mathbb{R}^n$,*

$$u^T v \le \|u\|_2 \|v\|_2.$$

If $v \neq 0$, equality holds exactly when u is a nonnegative multiple of v.

We will use one consequence over and over. Fix $v \neq 0$ and $\rho \ge 0$. Among all u with $\|u\|_2 \le \rho$, the inner product $u^T v$ is at most $\|u\|_2\|v\|_2 \le \rho\|v\|_2$, and the choice $u = \rho v/\|v\|_2$ attains this value. Hence

$$\sup_{\|u\|_2 \le \rho} u^T v = \rho\|v\|_2. \tag{1}$$

The formula also holds when $v = 0$, since then both sides are zero. In words: the vector in a ball that is most aligned with v points in the direction of v.

A similar argument, known as Hölder's inequality, works for the other pair of norms. Since $|u_i v_i| \le \|u\|_\infty |v_i|$ for each i,

$$u^T v \le \|u\|_\infty \|v\|_1, \qquad \text{so} \qquad \sup_{\|u\|_\infty \le \rho} u^T v = \rho\|v\|_1, \tag{2}$$

with the supremum attained at $u_i = \rho\,\mathbf{sign}(v_i)$. Here $\mathbf{sign}(v_i)$ is 1 if $v_i > 0$, -1 if $v_i < 0$, and any value in $[-1, 1]$ if $v_i = 0$.

Definition 2.2 (Support function)**.** The *support function* of a set $\mathcal{U} \subseteq \mathbb{R}^k$ is

$$\sigma_{\mathcal{U}}(v) = \sup_{u \in \mathcal{U}} u^T v.$$

Equation (1) says that the support function of the Euclidean ball of radius ρ is $\rho\|\cdot\|_2$, and (2) says that the support function of the box $\{u : \|u\|_\infty \le \rho\}$ is $\rho\|\cdot\|_1$. (If you remember Fenchel conjugates: $\sigma_{\mathcal{U}}$ is the conjugate of the indicator function of $\mathcal{U}$.)

2.2 Symmetric and positive semidefinite matrices

Let $\mathbf{S}^n$ denote the set of symmetric $n \times n$ real matrices. We equip $\mathbf{S}^n$ with the *trace inner product*

$$\langle X, Y \rangle = \mathbf{tr}(XY) = \sum_{i,j=1}^{n} X_{ij} Y_{ij}.$$

This is the ordinary dot product of the two matrices viewed as long vectors. Two identities are worth remembering: $\langle X, zz^T \rangle = z^T X z$, and $\langle X, I \rangle = \mathbf{tr}\, X$.

Eigenvalues. By the spectral theorem, every $X \in \mathbf{S}^n$ can be written as

$$X = Q \Lambda Q^T = \sum_{i=1}^{n} \lambda_i(X) q_i q_i^T,$$

where $Q = [q_1, \ldots, q_n]$ is orthogonal ($Q^T Q = I$) and Λ is diagonal, holding the real eigenvalues $\lambda_1(X) \ge \cdots \ge \lambda_n(X)$. We write $\lambda_{\max}(X) = \lambda_1(X)$ and $\lambda_{\min}(X) = \lambda_n(X)$. We always write the argument X in $\lambda_i(X)$, because the bare symbol λ is reserved for Lagrange multipliers.

Proposition 2.3 (Rayleigh quotient)**.** *For every $X \in \mathbf{S}^n$,*

$$\lambda_{\max}(X) = \max_{\|z\|_2=1} z^T X z, \qquad \lambda_{\min}(X) = \min_{\|z\|_2=1} z^T X z.$$

Proof. Write $z = Qw$; then $\|w\|_2 = \|z\|_2 = 1$ and $z^T X z = w^T \Lambda w = \sum_i \lambda_i(X) w_i^2$. This is a weighted average of the eigenvalues (the weights w_i^2 are nonnegative and sum to one), so it lies between $\lambda_{\min}(X)$ and $\lambda_{\max}(X)$. Choosing $z = q_1$ or $z = q_n$ attains the two extremes. □

Definition 2.4 (Positive semidefinite matrix)**.** A matrix $X \in \mathbf{S}^n$ is *positive semidefinite* (PSD), written $X \succeq 0$, if

$$z^T X z \geq 0 \qquad \text{for every } z \in \mathbb{R}^n.$$

It is *positive definite*, written $X \succ 0$, if $z^T X z > 0$ for every $z \neq 0$. We write $X \succeq Y$ to mean $X - Y \succeq 0$. The set of PSD matrices is denoted $\mathbf{S}^n_+$.

By Proposition 2.3, $X \succeq 0$ if and only if $\lambda_{\min}(X) \geq 0$, that is, all eigenvalues are nonnegative. Three further facts will be useful.

- *Gram matrices.* $X \succeq 0$ if and only if $X = VV^T$ for some matrix V. If v_i^T denotes row i of V, then $X_{ij} = v_i^T v_j$. (Given $X = Q\Lambda Q^T \succeq 0$, take $V = Q\Lambda^{1/2}$.)
- *Rank.* The rank of $X \succeq 0$ is its number of positive eigenvalues, and it equals the smallest number of columns a factor V can have. In particular, $X \succeq 0$ has rank one if and only if $X = xx^T$ for some $x \neq 0$.
- *Small principal submatrices.* If $X \succeq 0$, then every diagonal entry is nonnegative and every 2×2 principal submatrix is PSD (Exercise 2.5).

Exercise 2.5. Let $X \succeq 0$. Show that $X_{ii} \geq 0$ for every i and that $X_{ij}^2 \leq X_{ii} X_{jj}$ for every pair i, j. Conclude that if $X \succeq 0$ has unit diagonal, then $|X_{ij}| \leq 1$ for all i, j.

Exercise 2.6. Let $X = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. Find the eigenvalues and eigenvectors of X. Is $X \succeq 0$? Which unit vector z maximizes $z^T X z$?

3 Robust optimization and the second-order cone

We start with an application that produces a new kind of constraint all by itself.

3.1 A nominal production plan

A firm chooses amounts $x_1, x_2 \geq 0$ of two products. Product 1 earns profit 3 per unit and product 2 earns 2 per unit. Each unit of either product uses one unit of a shared resource, and one unit of the resource is available. The resulting LP is

$$\begin{array}{ll} \text{maximize} & 3x_1 + 2x_2 \\ \text{subject to} & x_1 + x_2 \leq 1 \\ & x_1, x_2 \geq 0. \end{array}$$

The optimal plan is $x = (1, 0)$, with profit 3: every unit of capacity goes to the more profitable product, and no capacity is left over.

Now suppose the resource use of each product is *estimated*. The true resource use is $a_1x_1 + a_2x_2$, and we only know that $a \approx (1, 1)$. If product 1 actually uses 10% more of the resource than expected, the plan $(1, 0)$ uses 1.1 units of capacity, and it is infeasible. The optimal plan of an LP typically sits at a vertex where constraints are tight, so it has no slack to absorb errors in the data. This is the motivation for robust optimization.

3.2 Uncertainty sets and robust constraints

Definition 3.1 (Ellipsoidal uncertainty)**.** We model an uncertain coefficient vector $a \in \mathbb{R}^n$ as

$$a = \bar{a} + Pu, \qquad u \in \mathcal{U} = \{u \in \mathbb{R}^k : \|u\|_2 \le \rho\},$$

where $\bar{a} \in \mathbb{R}^n$, $P \in \mathbb{R}^{n\times k}$, and $\rho \ge 0$ are given. The set $\mathcal{U}$ is the *uncertainty set*. The *nominal* model uses $a = \bar{a}$, that is, $u = 0$.

Each ingredient has a job.

- $\bar{a}$ is our best estimate of a.
- P describes the directions and relative sizes of the errors. For example, $P = \mathbf{diag}(0.5, 0.1)$ says that the error in a_1 can be five times as large as the error in a_2.
- ρ controls how much uncertainty we protect against. Larger ρ means more protection.

The set of possible vectors $\{\bar{a} + Pu : \|u\|_2 \le \rho\}$ is an *ellipsoid* centered at $\bar{a}$: the image of a ball under an affine map.

We must choose the decision x *before* we learn the error u. So we ask for one x that is feasible for every allowed error:

$$(\bar{a} + Pu)^T x \le b \qquad \text{for all } u \text{ with } \|u\|_2 \le \rho. \tag{3}$$

This is called a *robust constraint*. It is one linear inequality in x for each u in the ball, so it is an infinite family of linear constraints. Since an intersection of halfspaces is convex, the feasible set is convex; but we cannot hand infinitely many constraints to a solver. Our job is to find an equivalent finite description.

You have seen this situation once before. When the uncertainty set was a polyhedron, LP duality turned the "for all u" into finitely many linear constraints. Today the uncertainty set is a Euclidean ball, and the tool is the Cauchy–Schwarz inequality.

3.3 The worst error for a fixed decision

Fix a decision x. Constraint (3) holds for every u in the ball exactly when it holds for the *worst* u, the one that makes the left-hand side largest:

$$\bar{a}^T x + \sup_{\|u\|_2 \le \rho} u^T P^T x \le b.$$

To compute the supremum, set $v = P^T x$. By (1), the worst error is $u^\star = \rho v/\|v\|_2$, and the supremum equals $\rho\|v\|_2 = \rho\|P^T x\|_2$. Figure 1 shows the picture: the level lines of $u^T v$ are perpendicular to v, and the last one to touch the ball touches it at $u^\star$.

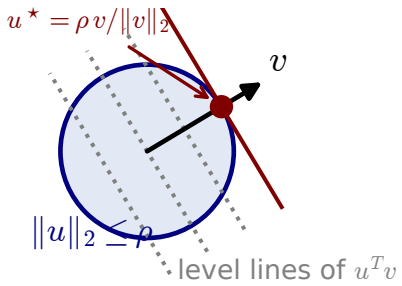


Figure 1: The worst error for a fixed decision. The linear function $u^T v$ increases in the direction $v = P^T x$; over the ball $\|u\|_2 \le \rho$ it is largest at $u^\star = \rho v/\|v\|_2$.

Proposition 3.2 (Robust counterpart for ellipsoidal uncertainty)**.** *A decision x satisfies the robust constraint* (3) *if and only if*

$$\bar{a}^T x + \rho\|P^T x\|_2 \le b. \tag{4}$$

Proof. The constraint holds for all u in the ball if and only if it holds at the supremum over the ball, and that supremum is $\bar{a}^T x + \rho\|P^T x\|_2$ by (1). □

We call (4) the *exact robust counterpart*. Read it as "nominal resource use plus a safety margin." The margin $\rho\|P^T x\|_2$ grows with the protection level ρ and with how much the plan x loads on the uncertain directions.

3.4 The second-order cone

The robust counterpart contains a Euclidean norm bounded by an affine function. That shape has a name.

Definition 3.3 (Second-order cone)**.** The *second-order cone* (SOC), also called the Lorentz or ice-cream cone, is

$$\mathcal{Q}^{k+1} = \{(z, t) \in \mathbb{R}^k \times \mathbb{R} : \|z\|_2 \le t\}.$$

Figure 2 shows $\mathcal{Q}^3$. For each height $t \ge 0$ the slice is a disk of radius t; for $t < 0$ the slice is empty.

Write the robust counterpart (4) as $\|\rho P^T x\|_2 \le b - \bar{a}^T x$. This says exactly that

$$(\rho P^T x,\ b - \bar{a}^T x) \in \mathcal{Q}^{k+1}.$$

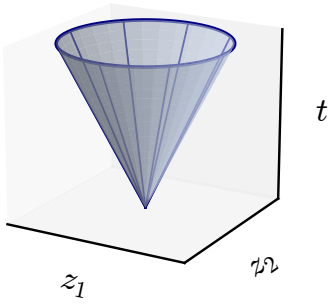


Figure 2: The second-order cone $\mathcal{Q}^3 = \{(z_1, z_2, t) : \sqrt{z_1^2 + z_2^2} \le t\}$.

Both entries of the pair are affine functions of x. So the infinitely many linear constraints in (3) have become *one* cone constraint on an affine expression.

Definition 3.4 (Second-order cone program)**.** A *second-order cone program* (SOCP) minimizes or maximizes a linear objective subject to affine equality constraints, linear inequality constraints, and constraints of the form

$$(F_i x + g_i,\ h_i^T x + d_i) \in \mathcal{Q}^{k_i+1}, \qquad \text{that is,} \qquad \|F_i x + g_i\|_2 \le h_i^T x + d_i.$$

Gotcha 3.5 (A negative right-hand side). If $b - \bar{a}^T x < 0$, then the SOC constraint is violated no matter what the norm term is, because a norm is never negative. This is the correct behavior: such a plan violates even the nominal constraint.

3.5 A worked example

Take $\bar{a} = (1, 1)$, $P = \mathbf{diag}(0.5, 0.1)$, and $\rho = 2$. Then $\rho P^T x = (x_1, 0.2x_2)$, and the robust production problem is the SOCP

$$\begin{array}{ll} \text{maximize} & 3x_1 + 2x_2 \\ \text{subject to} & x_1 + x_2 + 2\sqrt{(0.5x_1)^2 + (0.1x_2)^2} \le 1 \\ & x_1, x_2 \ge 0. \end{array}$$

A solver returns

$$x^\star \approx (0.1360,\ 0.6727), \qquad \text{profit} \approx 1.7534.$$

Let us check that the constraint is tight. The nominal use is $0.1360 + 0.6727 = 0.8087$, and the safety margin is $2\sqrt{0.0680^2 + 0.0673^2} \approx 2(0.0957) = 0.1913$, for a total of 1.0000.

Two observations help build intuition.

- The nominal plan $(1, 0)$ is badly exposed. Its robust resource use is $1 + 2\|(0.5, 0)\|_2 = 2$, twice the available capacity.
- The robust plan favors product 2, even though product 1 is more profitable. The reason is that product 1's resource use is five times as uncertain, so each unit of product 1 consumes much more of the safety margin.

Exercise 3.6. Use the same data with $\rho = 1$. Show that the plan $x = (2/3, 0)$ satisfies the robust constraint with equality, and compute its profit. Is it more or less profitable than the $\rho = 2$ plan? Why must that be so?

3.6 Reformulation, restriction, or relaxation?

When we replace one optimization problem by another, it helps to name the relationship.

Definition 3.7 (Exact reformulation, restriction, relaxation)**.** Consider an original problem with feasible set $\mathcal{X}$ and a new problem with the same objective and feasible set $\mathcal{X}'$ (possibly after a change of variables).

- The new problem is an *exact reformulation* if the two problems have the same optimal value and optimal solutions map back and forth between them.
- It is a *restriction* if $\mathcal{X}' \subseteq \mathcal{X}$. Its solutions are feasible for the original problem, but they may be conservative.
- It is a *relaxation* if $\mathcal{X}' \supseteq \mathcal{X}$. Its optimal value is a *bound* on the original optimal value: an upper bound when we maximize and a lower bound when we minimize.

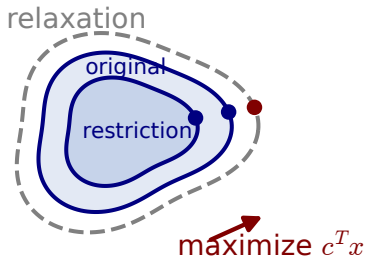


Figure 3: Nested feasible sets. A restriction shrinks the original feasible set; a relaxation enlarges it. When we maximize $c^T x$, the relaxation's optimum is at least as large as the original's, which is at least as large as the restriction's.

Where does our SOC constraint fit? It is an *exact reformulation* of the robust model, by Proposition 3.2. At the same time it is a *restriction* of the nominal model: any x that satisfies the robust constraint satisfies the nominal one (take $u = 0$). The answer depends on which model we compare with, so always say "exact for what?" We will meet an important relaxation when we study MaxCut in Section 7.

3.7 The price of protection

Figure 4 shows the feasible sets for several values of ρ. They are nested: if $\rho_1 \le \rho_2$, every plan that is protected against errors of size ρ_2 is also protected against errors of size ρ_1. Consequently the optimal profit can only decrease as ρ increases. Protection has a price, paid in nominal profit.

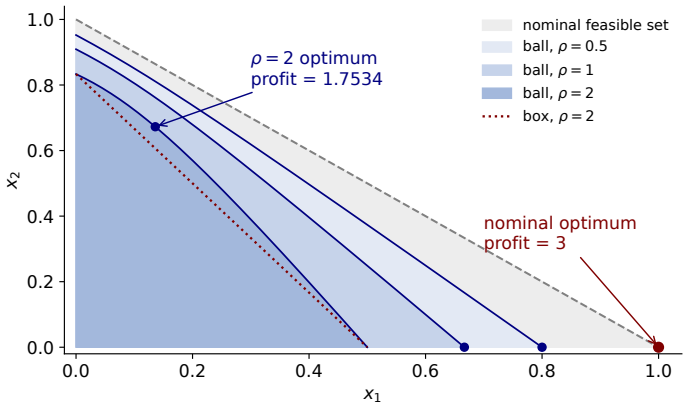


Figure 4: Feasible sets of the production problem: nominal, ellipsoidal uncertainty with $\rho = 0.5, 1, 2$, and box uncertainty with $\rho = 2$. Larger ρ gives a smaller feasible set and a lower optimal profit.

3.8 What protection buys: a probabilistic reading

The robust model protects against every error in a set; it says nothing about probabilities. But if we are willing to model the errors as random, we can measure what the protection buys. Let Φ denote the cumulative distribution function (CDF) of a standard normal random variable.

Suppose $u \sim \mathcal{N}(0, I)$. For a fixed plan x, the extra resource use $u^T P^T x$ is a linear combination of independent standard normals, so it is normal with mean zero and variance $\|P^T x\|_2^2$. The robust optimum leaves slack exactly $\rho\|P^T x\|_2$ in the nominal constraint (the robust constraint is tight). Hence

$$\mathbf{P}(\text{plan exceeds capacity}) = \mathbf{P}\big(u^T P^T x > \rho\|P^T x\|_2\big) = 1 - \Phi(\rho).$$

For the production example (see Figure 5):

- the nominal plan ($\rho = 0$) exceeds capacity with probability 50%;
- $\rho = 1.645$ gives probability 5% and profit about 1.82;
- $\rho = 2$ gives probability 2.3% and profit about 1.75.

The same calculation, run backwards, handles *chance constraints*. Suppose $a \sim \mathcal{N}(\bar{a}, PP^T)$ and we require the constraint to hold with probability at least η. Since $a^T x \sim \mathcal{N}(\bar{a}^T x, \|P^T x\|_2^2)$,

$$\mathbf{P}(a^T x \le b) \ge \eta \quad \Longleftrightarrow \quad \bar{a}^T x + \Phi^{-1}(\eta)\|P^T x\|_2 \le b.$$

When $\eta \ge \frac{1}{2}$, the coefficient $\Phi^{-1}(\eta)$ is nonnegative, and this is an SOC constraint: a robust constraint with $\rho = \Phi^{-1}(\eta)$. (When $\eta < \frac{1}{2}$ the coefficient is negative and the constraint is not convex.)

Exercise 3.8. Which value of ρ makes the ρ-robust plan exceed capacity with probability 1% when $u \sim \mathcal{N}(0, I)$? Use $\Phi^{-1}(0.99) \approx 2.326$.

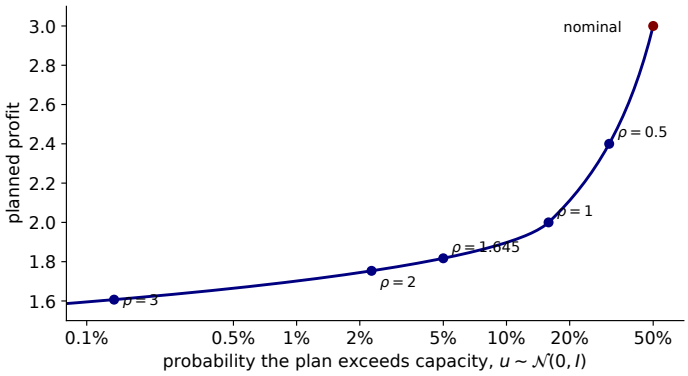


Figure 5: Planned profit versus the probability that the plan exceeds capacity, when $u \sim \mathcal{N}(0, I)$. Each point is a robust optimum for one value of ρ.

3.9 The uncertainty set is a modeling choice

The ball is not the only reasonable uncertainty set. The worst extra resource use is always the support function $\sigma_{\mathcal{U}}(P^T x)$, and different sets give different counterparts:

allowed errors	worst extra resource use	robust counterpart
$\|u\|_2 \le \rho$	$\rho \|P^T x\|_2$	SOCP
$\|u\|_\infty \le \rho$	$\rho \|P^T x\|_1$	LP
$Du \le d$	$\min_\eta \{d^T \eta : D^T \eta = P^T x,\ \eta \ge 0\}$	LP

The box row uses (2). Its counterpart is an LP because $\|P^T x\|_1 \le \tau$ can be written with linear inequalities (introduce $s \ge P^T x$, $s \ge -P^T x$, and require $\bar{a}^T x + \rho \mathbf{1}^T s \le b$).

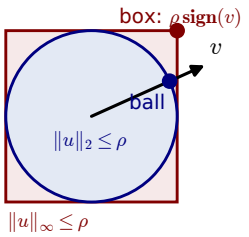


Figure 6: For the same ρ, the box $\|u\|_\infty \le \rho$ contains the ball $\|u\|_2 \le \rho$, so the box model protects against more errors.

For the same P and ρ, the box model is more conservative, since the box contains the ball (Figure 6). Two cautions apply to every choice.

- Choose P and ρ from domain knowledge and data, and validate the plan out of sample.
- Protection holds only for errors *inside* the chosen set. For instance, $\rho = 2$ implies a 97.7% guarantee only under a Gaussian model; with other error distributions, ρ must be calibrated.

Polyhedral uncertainty. Let $\mathcal{U} = \{u : Du \le d\}$ be nonempty and bounded. By LP strong duality,

$$\sup_{u \in \mathcal{U}} v^T u = \min_{\eta \ge 0} \{d^T \eta : D^T \eta = v\}.$$

Hence the robust constraint $\bar{a}^T x + \sup_{u \in \mathcal{U}} u^T P^T x \le b$ holds if and only if there exists η with

$$\eta \ge 0, \qquad D^T \eta = P^T x, \qquad \bar{a}^T x + d^T \eta \le b.$$

An existential variable is enough to enforce a constraint for every u: any dual feasible η gives an upper bound on the worst case (weak duality), and strong duality makes the best such bound exact. Each robust row can use its own η.

Exercise 3.9. Solve the production problem with box uncertainty: $\bar{a} = (1, 1)$, $P = \mathbf{diag}(0.5, 0.1)$, and $\|u\|_\infty \le 2$. (Hint: the robust counterpart is a single linear inequality in $x \ge 0$. Compare the optimal profit with the ball model's 1.7534.)

4 Why expose conic structure?

We now have a convex constraint, $\bar{a}^T x + \rho \|P^T x\|_2 \le b$. How should we give it to a solver? There are at least three ways to write it.

1. *As written.* The function $x \mapsto \|P^T x\|_2$ is convex but not differentiable where $P^T x = 0$. Many solvers assume differentiable functions.

2. *Squared.* Since both sides are nonnegative on the feasible set, we can write

   $$\rho^2 \|P^T x\|_2^2 \le (b - \bar{a}^T x)^2, \qquad b - \bar{a}^T x \ge 0.$$

   Now every function is smooth. But the function $\rho^2 \|P^T x\|_2^2 - (b - \bar{a}^T x)^2$ is a difference of convex functions and is not convex, so convexity is hidden from the solver. Moreover, at the tip of the cone ($P^T x = 0$ and $b = \bar{a}^T x$) the gradient of this function is zero, so a linearization of the constraint carries no information there.

3. *As a cone.* $(\rho P^T x,\ b - \bar{a}^T x) \in \mathcal{Q}^{k+1}$: an affine expression constrained to lie in a standard convex set.

Nonsmooth convex functions are common, not exotic. Examples include ℓ_1 penalties (which promote sparsity), maxima of affine functions, and the largest eigenvalue $\lambda_{\max}$ of a symmetric matrix. In facility location, we choose a site x to minimize the total distance $\sum_i \|x - p_i\|_2$ to customers p_i, and the optimum often lies exactly at a kink, $x = p_i$ (Figure 7).

4.1 Two solver interfaces

Solvers differ in what they need to be told about a problem.

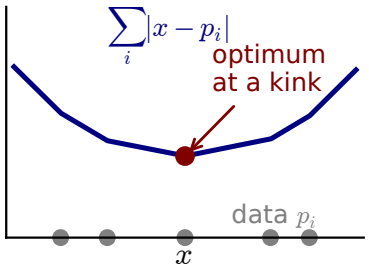


Figure 7: A one-dimensional facility location objective $\sum_i |x - p_i|$. The optimum sits at a data point, exactly where the objective is not differentiable.

Derivative-oracle solvers. A solver such as IPOPT asks the user (or a modeling tool) to supply, at any point x it requests, the objective value $f(x)$, the constraint values $g(x)$, the gradient $\nabla f(x)$, the constraint Jacobian $J_g(x)$, and often the Hessian of the Lagrangian $\nabla^2_{xx}\mathcal{L}$. A routine that returns such information on request is called an *oracle*. The derivatives are often computed by *automatic differentiation*, which applies the chain rule to the code that evaluates the function. These solvers assume twice-differentiable functions, and they handle flexible smooth models, including nonconvex ones. They seek a *KKT point*: a point satisfying the Karush–Kuhn–Tucker conditions (feasibility, stationarity of the Lagrangian, and complementary slackness).

Conic solvers. A solver such as Clarabel or MOSEK asks for data A, b, c and a list of standard cone types. The user provides affine maps and cone constraints, and the solver has specialized routines for each cone type. Because every standard cone is convex, convexity is built into the input form.

Both IPOPT and Clarabel use *interior-point methods*, a family of algorithms that we will study after the midterm. For now, think of them as methods that take a modest number of Newton-like steps through the interior of the feasible region.

4.2 What conic structure buys, and what it costs

A conic formulation offers four advantages.

- *Nonsmooth functions, exactly.* Norms, absolute values, and maxima become affine maps plus cones. No solver ever needs the derivative of a kink.
- *Global answers with certificates.* A *certificate* is a mathematical proof that you can check independently of the solver. A conic solver returns either a primal–dual pair whose duality gap proves optimality, or a proof of infeasibility. We develop both in Section 8.
- *Constraints with no good smooth form.* The constraint $X \succeq 0$ and the function $\lambda_{\max}$ are hard to express smoothly: eigenvalues are not differentiable where two of them coincide.
- *Predictable work.* Interior-point iteration counts barely grow with problem size.

There is a tradeoff. Conic formulations often introduce extra variables and constraints, which can enlarge the numerical work, and a compact smooth model can be very effective. In particular, two mathematically equivalent formulations need not take the same time to solve.

4.3 When to use which solver

Use a *conic solver* (Clarabel or MOSEK, usually through a modeling language such as CVXPY) when the problem is convex and built from norms, quadratics, logarithms, or matrix inequalities; when it has nonsmooth terms such as ℓ_1 norms, maxima, or $\lambda_{\max}$; or when you need certificates or infeasibility detection.

Use a *derivative-oracle solver* (IPOPT, KNITRO, or SNOPT) for smooth *nonconvex* models, such as power flow, chemical processes, or models with ODE dynamics; for functions given by simulation code or a neural network; or for a compact smooth model that a conic formulation would enlarge.

For very large problems with cheap gradients, *first-order methods*, which use only gradients and no second derivatives, are often the right choice. We study them after the midterm.

Gotcha 4.1 (A KKT point of a convex problem). If IPOPT returns a KKT point of a convex problem, that point is globally optimal: for convex problems, the KKT conditions are sufficient. The weaknesses of the derivative-oracle approach lie elsewhere. It cannot *certify* infeasibility, and kinks violate its smoothness assumptions.

5 Conic form

5.1 Convex cones

Definition 5.1 (Convex cone)**.** A nonempty set K is a *convex cone* if

$$x, y \in K, \quad \alpha, \beta \geq 0 \quad \Longrightarrow \quad \alpha x + \beta y \in K.$$

Taking $\alpha = \beta = 0$ shows that every convex cone contains 0. Taking $\beta = 0$ shows that a cone contains every nonnegative multiple of each of its points: it is a union of rays from the origin. Taking $\alpha + \beta = 1$ shows that a convex cone is a convex set.

The following cones are the building blocks of modern solvers.

cone	membership	models
$\{0\}$ (zero cone)	$z = 0$	equalities
$\mathbb{R}^m_+$ (nonnegative orthant)	$z \geq 0$	linear inequalities
$\mathcal{Q}^{m+1}$ (second-order cone)	$\|z\|_2 \leq t$	norm bounds, quadratics
$\mathbf{S}^m_+$ (PSD cone)	$Z \succeq 0$	eigenvalue bounds
$K_{\exp}$ (exponential cone)	$ye^{x/y} \leq z,\ y > 0$ (closure)	exp, log, entropy

Exercise 5.2. Use the triangle inequality to show that $\mathcal{Q}^{m+1}$ is a convex cone. Then show directly from the definition of PSD that $\mathbf{S}^m_+$ is a convex cone.

Cones beyond SDP. The *exponential cone* is the closure of $\{(x, y, z) : y > 0,\ ye^{x/y} \leq z\}$. Taking $y = 1$ gives $e^x \leq z$, so it represents epigraphs of exponentials and hypographs of logarithms, as well as entropy and the logistic loss. The *power cone* $\{(x, y, z) : x^\alpha y^{1-\alpha} \geq |z|,\ x, y \geq 0\}$, for $\alpha \in (0, 1)$, represents powers $|x|^p$ and p-norms. Modern conic solvers accept

all of these cones, mixed freely in one problem.

5.2 Conic programs

Definition 5.3 (Conic program)**.** A *conic program* has the form

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & b - Ax \in K, \end{array}$$

where $x \in \mathbb{R}^n$ is the variable, $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $c \in \mathbb{R}^n$ are data, and $K \subseteq \mathbb{R}^m$ is a closed convex cone. Constraints on matrices use linear maps in place of A.

The expression $b - Ax$ is sometimes called the *slack*, and we write $s = b - Ax$. For example, the LP constraint $Ax \leq b$ is the conic constraint $b - Ax \in \mathbb{R}^m_+$: the slack must be nonnegative.

A real problem usually has several kinds of constraints. We combine them with a *product cone*:

$$z_1 \in K_1,\ z_2 \in K_2 \quad \Longleftrightarrow \quad (z_1, z_2) \in K_1 \times K_2.$$

A product of convex cones is a convex cone, so this costs nothing. The cone types that appear give the problem its name:

- zero cones and nonnegative orthants only: a *linear program* (LP);
- add second-order cones: a *second-order cone program* (SOCP);
- add PSD cones: a *semidefinite program* (SDP).

Gotcha 5.4 (An arbitrary cone is not an easy problem)*.* Every convex problem can be written in conic form for *some* convex cone. That does not make the problem easy. Conic form is useful because solvers have efficient algorithms for a few *specific* cones, and many problems have manageable representations using those cones.

5.3 A hierarchy of conic programs

A *quadratic program* (QP) minimizes a convex quadratic objective $\frac{1}{2}x^T Hx + g^T x$, with $H \succeq 0$, subject to affine constraints. The classes we have named are nested (Figure 8):

$$\text{LP} \subset \text{QP} \subset \text{SOCP} \subset \text{SDP}.$$

Each inclusion is an exact reformulation.

- LP $\subset$ QP: take $H = 0$.
- QP $\subset$ SOCP: an epigraph variable plus the quadratic-epigraph representation in Exercise 5.11.
- SOCP $\subset$ SDP: the matrix representation of the SOC in Proposition 6.4 below.

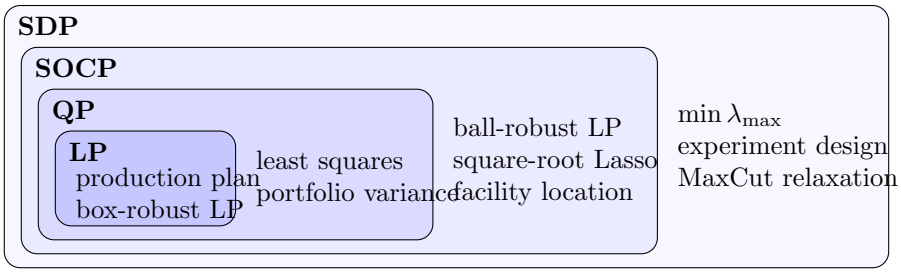


Figure 8: The hierarchy of conic programs, with examples from these notes placed in the smallest class that contains them.

If SDP contains everything in the picture, why not always formulate an SDP? Because more general cones cost more per iteration. A solver that knows a constraint is an SOC can exploit that structure; we will see numbers in Section 6.

The exponential cone lies *outside* this picture: no SDP represents exp or log exactly. (Sets representable by SDPs are described by polynomial inequalities, and the exponential function is not.) Solvers simply mix all these cones in one problem.

5.4 The epigraph trick and conic representations

Definition 5.5 (Epigraph)**.** The *epigraph* of a function f is the set of points on or above its graph:

$$\mathbf{epi}\, f = \{(x,t) : f(x) \le t\}.$$

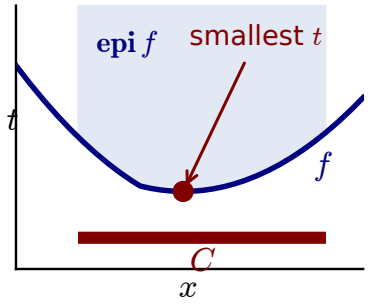


Figure 9: Minimizing f over C is the same as finding the lowest point of the epigraph above C: the smallest t with $f(x) \le t$.

Proposition 5.6 (Epigraph reformulation)**.** *For any function f and set C, the problems*

$$\underset{x\in C}{\text{minimize}}\ f(x) \qquad \text{and} \qquad \begin{array}{ll}\text{minimize}_{x,t} & t \\ \text{subject to} & f(x) \le t, \quad x \in C\end{array}$$

have the same optimal value, and $x^\star$ solves the first if and only if $(x^\star, f(x^\star))$ solves the second.

Proof. Every feasible x for the first problem lifts to the feasible point $(x, f(x))$ of the second, with the same objective value. Conversely, every feasible (x, t) of the second problem has objective $t \geq f(x)$, so it is no better than the point x in the first problem. Hence the two optimal values agree. □

Notice that the proof never used convexity: the epigraph trick is always exact. Convexity enters elsewhere: the epigraph is a convex set exactly when f is a convex function.

Definition 5.7 (Conic representation)**.** A *conic representation* of a function f expresses its epigraph using affine maps, standard cones, and possibly extra variables: $f(x) \leq t$ holds if and only if there exists w such that (x, t, w) satisfies a set of conic constraints.

Here are three representations we will use.

- *Euclidean norm:* $\|z\|_2 \leq t \iff (z, t) \in \mathcal{Q}^{m+1}$. No extra variables.
- *ℓ_1 norm:* $\|x\|_1 \leq t \iff \exists s : \ -s \leq x \leq s, \ \mathbf{1}^T s \leq t$. The extra variable s bounds $|x_i|$ componentwise.
- *Maximum of affine functions:* $\max_i(a_i^T x + b_i) \leq t \iff a_i^T x + b_i \leq t$ for each i.

Gotcha 5.8 (The direction of the inequality matters). The representation $\|x\|_1 \leq 1 \iff \exists s : -s \leq x \leq s, \ \mathbf{1}^T s \leq 1$ is correct. What if we reverse both ≤ 1 inequalities, hoping to represent $\|x\|_1 \geq 1$? The proposed formulation is

$$\exists s : \ -s \leq x \leq s, \ \mathbf{1}^T s \geq 1.$$

Try $x = 0$. Any $s \geq 0$ with $\mathbf{1}^T s = 1$ satisfies all the constraints, so $x = 0$ is "feasible," yet $\|0\|_1 = 0 \not\geq 1$. The extra variable s is only an *upper* bound on $|x|$, and an upper bound can be made as large as we like; so it cannot enforce a *lower* bound on $\|x\|_1$. (Indeed $\|x\|_1 \geq 1$ is a nonconvex constraint, so no conic representation exists.)

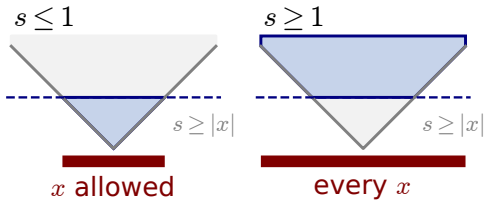


Figure 10: The case $n = 1$. Left: the lifted set $\{(x, s) : |x| \leq s \leq 1\}$ projects onto the allowed set $|x| \leq 1$. Right: the lifted set $\{(x, s) : |x| \leq s, \ s \geq 1\}$ projects onto every x, because s can always be chosen large.

5.5 A complete example: square-root Lasso

The *square-root Lasso* is a regression method that trades off residual size against the sparsity of the coefficients:

$$\underset{x\in\mathbb{R}^n}{\text{minimize}} \quad \|Ax-b\|_2+\gamma\|x\|_1,$$

where $A\in\mathbb{R}^{m\times n}$, $b\in\mathbb{R}^m$, and $\gamma>0$. (We call the parameter γ rather than λ so that λ can denote Lagrange multipliers later.) Before reading on, try to transform this problem into conic form yourself. Ask: what extra variables do you need, which constraints use which cones, and why is the reformulation exact?

Here is one answer. Introduce $s\in\mathbb{R}^n$ to bound $|x|$, a residual $r\in\mathbb{R}^m$, and a scalar t to bound $\|r\|_2$:

$$\begin{array}{lll} \text{minimize}_{x,s,r,t} & t+\gamma\mathbf{1}^Ts & \\ \text{subject to} & s-x\geq 0, \quad s+x\geq 0 & \text{cone } \mathbb{R}^n_+\times\mathbb{R}^n_+ \\ & r-(Ax-b)=0 & \text{cone } \{0\}^m \\ & \|r\|_2\leq t & \text{cone } \mathcal{Q}^{m+1}. \end{array}$$

Why is it exact? We argue as in Proposition 5.6, in both directions.

- Every x lifts to a feasible point with the same objective: take $s=|x|$ (componentwise), $r=Ax-b$, and $t=\|r\|_2$. Then $t+\gamma\mathbf{1}^Ts=\|Ax-b\|_2+\gamma\|x\|_1$.
- Every feasible lifted point is no better than its x: since $s\geq|x|$ and $t\geq\|r\|_2=\|Ax-b\|_2$,

$$\|Ax-b\|_2+\gamma\|x\|_1\leq t+\gamma\mathbf{1}^Ts.$$

At an optimum the inequalities $s\geq|x|$ and $t\geq\|r\|_2$ are tight. Why? Both t and s appear in the minimized objective with positive coefficients, so if either bound had slack, we could decrease it and lower the objective.

What did lifting change? The original problem has n variables; the conic formulation has $2n+m+1$. In the form $b-Ax\in K$, all the constraints read

$$\begin{pmatrix} s-x \\ s+x \\ r-Ax+b \\ r \\ t \end{pmatrix} \in \mathbb{R}^n_+\times\mathbb{R}^n_+\times\{0\}^m\times\mathcal{Q}^{m+1}.$$

The objective and every expression inside a cone are affine in (x,s,r,t). We could also eliminate r and write the cone constraint as $(Ax-b,t)\in\mathcal{Q}^{m+1}$. The optimal value does not change, but the size and sparsity pattern of the data do. Equivalent formulations can have different sizes.

Exercise 5.9. Write $\underset{x}{\text{minimize}}\ \|Ax-b\|_\infty$ in conic form. Which cones appear? Explain why your reformulation is exact.

Exercise 5.10. Let $F\in\mathbb{R}^{k\times n}$, $g\in\mathbb{R}^k$, $h\in\mathbb{R}^n$, and $d\in\mathbb{R}$. Write the constraint $\|Fx+g\|_2\leq h^Tx+d$ as membership of an affine expression in a second-order cone. Then write the constraint $\|x\|_2\leq 1$ in the form $b-Ax\in\mathcal{Q}^{n+1}$: what are A and b?

Exercise 5.11. Let $z \in \mathbb{R}^m$ and $t \in \mathbb{R}$. Show that

$$\|z\|_2^2 \le t \quad \Longleftrightarrow \quad \left\| \begin{pmatrix} 2z \\ t-1 \end{pmatrix} \right\|_2 \le t+1.$$

Then explain how to represent the convex quadratic constraint $x^T H x \le t$, where $H = R^T R \succeq 0$, with a second-order cone.

6 From a model to a solver

6.1 What each solver receives

Let us give the square-root Lasso to two solvers. Both minimize $t + \gamma \mathbf{1}^T s$ over the same variables (x, s, r, t), subject to $-s \le x \le s$ and $r = Ax - b$. They differ only in how the norm bound is stated.

IPOPT	Clarabel
$r^T r \le t^2, \quad t \ge 0$	$(r, t) \in \mathcal{Q}^{m+1}$
a smooth scalar constraint and its derivatives	a standard cone constraint

Why does the scalar formulation need $t \ge 0$? Without it, a negative t with $t^2 \ge r^T r$ would satisfy the squared inequality, and the objective, which wants t small, would happily choose $t = -\|r\|_2$.

Both descriptions define the same convex feasible set. But the scalar function $r^T r - t^2$ is *not* convex (its Hessian has a negative eigenvalue). Notice also that to give IPOPT a smooth problem, *we did the conic lifting ourselves*: we introduced s, r, and t exactly as in the conic form.

6.2 Measured solve times

Table 1 reports one experiment. The matrix A is dense with Gaussian entries, and b comes from a sparse signal plus noise. Times are medians of three runs on a laptop, using one thread. IPOPT was called through CasADi with exact derivatives, and Clarabel through its native interface. "Setup" includes model construction and solver initialization. Every solve passed the same independent checks: relative feasibility violation and primal–dual gap at most 10^{-6}.

	solve only (ms)		setup + solve (ms)		iterations	
(m, n)	IPOPT	Clarabel	IPOPT	Clarabel	IPOPT	Clarabel
$(100, 50)$	40.9	6.4	50.8	12.2	24	13
$(300, 150)$	168.1	83.6	208.6	103.3	25	12
$(600, 300)$	873.7	610.6	1017.3	677.7	26	15

Table 1: Square-root Lasso solve times and iteration counts for three problem sizes.

Clarabel is faster here, and its iteration count is nearly flat as the problem grows. Treat this as an illustration, not a law: the advantage depends on the problem and on the formulation.

6.3 What if we skip the lifting?

Suppose instead that we hand IPOPT the original objective $f(x) = \|Ax - b\|_2 + \gamma \|x\|_1$, still with exact automatic derivatives. On the same 9 instances, with tolerance 10^{-9} and a limit of 500 iterations:

- IPOPT on the raw objective converged on 0 of the 9 instances; its final objective was 0.8–11.4% above optimal;
- Clarabel on the lifted model solved all 9 in at most 15 iterations, certified to 10^{-6}.

Why does IPOPT struggle here when it did fine on the lifted model? The function $|x_i|$ has no derivative at 0, which is exactly where a sparse solution puts most of its coordinates. IPOPT builds a local quadratic model from derivatives, and that model is wrong at a kink; many coordinates hover near zero instead of reaching it. The lifting is the useful work, and a conic solver accepts the lifted model natively.

6.4 Disciplined convex programming

How does software know that a model is convex? Consider two expressions for the same function:

$$f(x) = \sqrt{1 + x^2} = \left\| \begin{pmatrix} 1 \\ x \end{pmatrix} \right\|_2 .$$

On the left, f is the square root (a concave, increasing function) of the convex function $1 + x^2$. The usual composition rules say nothing about a concave increasing function of a convex function, so this route is inconclusive. On the right, f is a norm of an affine expression, which is convex. The function is the same; only the second expression comes with an easy proof.

Definition 6.1 (Disciplined convex programming)**.** *Disciplined convex programming* (DCP) builds expressions from a library of functions with known curvature (convex, concave, or affine) and monotonicity, combined using composition rules such as:

- a convex nondecreasing function of a convex expression is convex;
- a convex nonincreasing function of a concave expression is convex;
- a convex function of an affine expression is convex;

together with the mirror-image rules for concave functions, and the facts that nonnegative sums of convex expressions are convex and the negative of a convex expression is concave. A DCP problem has one of the forms

$$\text{minimize convex}, \qquad \text{maximize concave},$$

with constraints of the forms

$$\text{convex} \leq \text{concave}, \qquad \text{affine} = \text{affine}.$$

Gotcha 6.2 (A failed DCP check is not a proof of nonconvexity)*.* DCP rules are *sufficient* for convexity, not necessary. Software such as CVXPY may reject $\sqrt{1 + x^2}$ and accept $\|(1, x)\|_2$. If a DCP check fails, you may need a different expression for the same function, or a different formulation of the problem.

Exercise 6.3. Which of the following expressions are DCP-convex, using the rules above? For any that are not, decide whether the function is nonetheless convex and, if so, give a DCP expression for it. The variables are $x \in \mathbb{R}$ and $y \in \mathbb{R}^2$.

(a) $\exp(\|y\|_2)$.

(b) $-\sqrt{x}$, for $x \geq 0$.

(c) $\sqrt{\|y\|_2}$.

(d) $\sqrt{x^2 + 4}$.

6.5 What does the modeling language do?

A modeling language such as CVXPY checks convexity with DCP and then translates the model into a solver's input form. For the square-root Lasso:

```
x = cp.Variable(n)
model = cp.Problem(cp.Minimize(
    cp.norm(A @ x - b, 2) + gamma * cp.norm1(x)))
data, _, _ = model.get_problem_data(cp.CLARABEL)
print(data["A"].shape)   # size of the conic data matrix
print(data["dims"])      # how many rows belong to each cone
model.solve(solver=cp.CLARABEL)
```

Three things happen.

- *Canonicalization*: the modeling language introduces auxiliary variables and standard cone constraints, just as we did by hand.
- The solver computes primal and dual variables for the resulting conic problem.
- The modeling language maps the results back to the user's variables.

CVXPY's auxiliary variables need not match ours exactly, and they do not need to: any exact conic representation gives the same optimal x.

6.6 Choosing a formulation

The same constraint can often be written with different cones. Here is the key example behind the inclusion SOCP $\subset$ SDP.

Proposition 6.4 (The second-order cone as a matrix inequality)**.** *For $z \in \mathbb{R}^m$ and $t \in \mathbb{R}$,*

$$(z, t) \in \mathcal{Q}^{m+1} \quad \Longleftrightarrow \quad \begin{bmatrix} tI_m & z \\ z^T & t \end{bmatrix} \succeq 0.$$

Proof. We use the *Schur complement* fact: if $A \succ 0$, then

$$\begin{bmatrix} A & B \\ B^T & C \end{bmatrix} \succeq 0 \quad \Longleftrightarrow \quad C - B^T A^{-1} B \succeq 0.$$

(To see why, minimize the quadratic form $u^T A u + 2u^T B w + w^T C w$ over u for fixed w; the minimum is at $u = -A^{-1}Bw$ and equals $w^T(C - B^T A^{-1} B)w$.)

- If $t > 0$, take $A = tI$, $B = z$, $C = t$. The matrix is PSD if and only if $t - \|z\|_2^2/t \geq 0$, that is, $\|z\|_2 \leq t$.
- If $t = 0$, the matrix is $\begin{bmatrix} 0 & z \\ z^T & 0 \end{bmatrix}$. By Exercise 2.5, each 2×2 principal submatrix $\begin{bmatrix} 0 & z_i \\ z_i & 0 \end{bmatrix}$ must have $z_i^2 \leq 0$, so $z = 0$, which is exactly $\|z\|_2 \leq 0$.
- If $t < 0$, the diagonal entries are negative, so the matrix is not PSD; and $\|z\|_2 \leq t$ fails too.

□

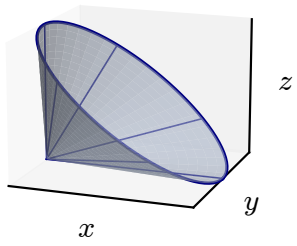


Figure 11: The 2×2 PSD matrices $\left[\begin{smallmatrix} x & y \\ y & z \end{smallmatrix}\right]$, drawn in coordinates (x, y, z). In rotated coordinates this set is a second-order cone.

Both sides describe the same set, but they are not equally cheap. The SOC form uses a cone of dimension $m + 1$; the PSD form uses a matrix of order $m + 1$, with about $m^2/2$ distinct entries. A solver that is told "this is an SOC" can exploit that structure directly. A more general representation does not give a faster method: preserve useful structure, and compare size, sparsity, and solver support.

Evidence: preserving quadratic structure. The Clarabel authors compared two ways of solving the 138 convex QPs of the Maros–Mészáros test set, a standard benchmark: keeping the quadratic objective $\frac{1}{2}x^T Hx + g^T x$ natively, or lifting it to an SOCP with an epigraph variable, $\min\{t + g^T x : \frac{1}{2}x^T Hx \leq t\}$. Their results (Goulart and Chen, 2026, Figure 2; solver-specific default tolerances) were:

138 QPs	normalized time	failure rate
Clarabel, native quadratic objective	1.00	10.9%
Clarabel, SOCP reformulation	11.82	43.5%
ECOS (another conic solver), SOCP reformulation	11.73	43.5%

Time is the shifted geometric mean, normalized so the best solver has time 1, with each failure charged 300 seconds. The mathematically equivalent SOCP was about twelve times slower and failed four times as often. Exact equivalence does not mean equal computational cost.

7 Semidefinite programs and relaxations

7.1 A bound in every direction

Definition 7.1 (Semidefinite program)**.** A *semidefinite program* (SDP) minimizes or maximizes a linear function subject to affine constraints and *linear matrix inequalities*: constraints of the form $F(x) \succeq 0$, where $F(x) = F_0 + \sum_i x_i F_i$ is a symmetric matrix that depends affinely on the variables. Equivalently, it is a conic program whose cone is a product of zero cones, nonnegative orthants, second-order cones, and PSD cones.

Why are matrix inequalities so expressive? By definition,

$$X \succeq 0 \quad \iff \quad z^T X z \geq 0 \quad \text{for every } z.$$

So one matrix inequality is an infinite family of scalar inequalities, one for each direction z. Each of them, $\langle X, zz^T \rangle \geq 0$, is *linear* in X, so the PSD cone is an intersection of halfspaces and is convex. Applying the definition to $tI - X$,

$$tI - X \succeq 0 \quad \iff \quad z^T X z \leq t\|z\|_2^2 \quad \text{for all } z \quad \iff \quad \lambda_{\max}(X) \leq t, \tag{5}$$

where the last step is Proposition 2.3. Each inequality in the middle is linear in the pair (X, t), so the set $\{(X, t) : \lambda_{\max}(X) \leq t\}$ is convex. This is a direct proof that $\lambda_{\max}$ is a convex function. In the same way, $\lambda_{\min}(X) \geq t$ if and only if $X - tI \succeq 0$, and $\lambda_{\min}$ is concave. An SDP encodes bounds on a quadratic form in *every* direction at once (Figure 12).

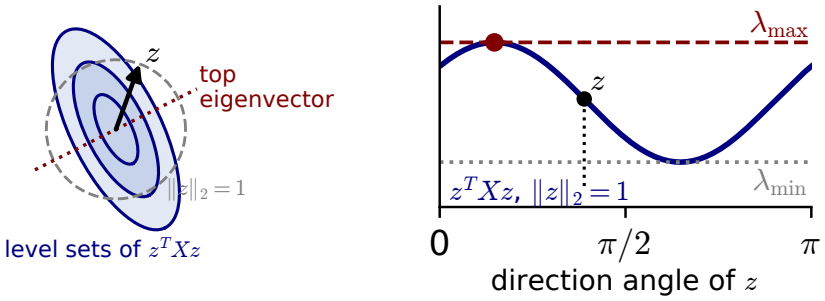


Figure 12: Left: level sets of a quadratic form $z^T X z$; it grows fastest along the top eigenvector. Right: $z^T X z$ for unit vectors z, as a function of the direction angle. Its maximum is $\lambda_{\max}$ and its minimum is $\lambda_{\min}$; the constraint $tI - X \succeq 0$ puts the horizontal line t above the whole curve.

7.2 Designing for the worst direction

Let $X(x) = X_0 + \sum_{i=1}^{n} x_i X_i$, where each $X_i \in \mathbf{S}^m$ is given. We want to choose x to make the largest eigenvalue of $X(x)$ as small as possible. By the epigraph trick and (5),

$$\underset{x \in C}{\text{minimize}}\ \lambda_{\max}(X(x)) \quad \Longleftrightarrow \quad \begin{array}{ll} \text{minimize}_{x,t} & t \\ \text{subject to} & tI - X(x) \succeq 0, \quad x \in C. \end{array}$$

The constraint $tI - X(x) \succeq 0$ is affine in (x, t) inside a PSD cone, so this is an SDP whenever C is described by affine and PSD constraints. Two points deserve emphasis. The reformulation is *exact*, not a relaxation. And $X(x)$ itself need not be PSD; it can be any symmetric matrix. Only $tI - X(x)$ is constrained.

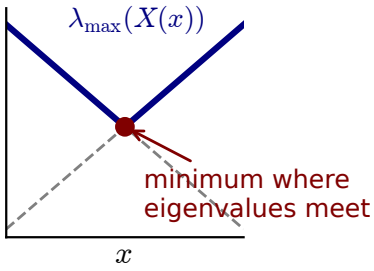


Figure 13: $\lambda_{\max}(X(x))$ for a one-parameter family. The minimum typically occurs where two eigenvalues meet, which is exactly where $\lambda_{\max}$ is not differentiable.

Exercise 7.2. Let $X(x) = \begin{bmatrix} x & 1 \\ 1 & -x \end{bmatrix}$ for $x \in \mathbb{R}$. Compute $\lambda_{\max}(X(x))$, find its minimizer, and write the SDP whose optimal value equals $\min_x \lambda_{\max}(X(x))$. Which function from Section 6 have you found?

7.3 Experiment design: where should we measure?

Here is an application where the worst direction has a physical meaning. We want to locate an unknown point $\theta \in \mathbb{R}^2$. We have six candidate sensors; sensor i returns a reading

$$y = v_i^T \theta + \varepsilon,$$

where $v_i \in \mathbb{R}^2$ is known and the noise $\varepsilon \sim \mathcal{N}(0, 1)$ is independent across readings. (A noisier sensor is modeled by a shorter v_i: dividing a reading by its noise level rescales v_i.) We can take N readings in total and must decide how to split them.

Let $w_i \geq 0$ be the fraction of the N readings spent on sensor i, so $\mathbf{1}^T w = 1$. Define the *information matrix*

$$M(w) = \sum_{i=1}^{6} w_i v_i v_i^T.$$

The least-squares estimate $\hat{\theta}$ from the N readings has error $\hat{\theta} - \theta \sim \mathcal{N}\big(0, \tfrac{1}{N} M(w)^{-1}\big)$. (This is the familiar covariance $(V^T V)^{-1}$ of least squares, where the rows of V are the measurement vectors; here $V^T V = \sum_k v_{i_k} v_{i_k}^T = N M(w)$.)

The *error ellipse* $\{\delta : \delta^T M(w)\delta \le 1\}$ describes the shape of the uncertainty. Its axes point along the eigenvectors of $M(w)$, and the semi-axis along an eigenvector with eigenvalue μ has length $1/\sqrt{\mu}$. So we learn θ well along directions we measure often, and poorly along directions we neglect. The worst-known direction is the eigenvector for $\lambda_{\min}(M(w))$, the long axis of the ellipse.

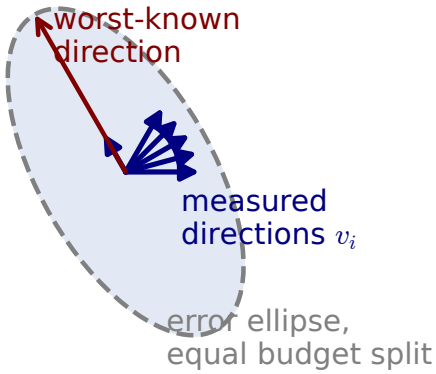


Figure 14: Six candidate sensor directions v_i: five similar ones, and one noisier (shorter) sensor pointing in a new direction. With an equal budget split, the error ellipse is long in the poorly measured direction.

E-optimal design. To shrink the longest axis of the error ellipse, we minimize $\lambda_{\max}(M(w)^{-1}) = 1/\lambda_{\min}(M(w))$; equivalently, we maximize $\lambda_{\min}(M(w))$. The matrix $M(w)$ is *affine* in w (it is linear in w), and $\lambda_{\min}(M) \ge t$ if and only if $M - tI \succeq 0$. So the *E-optimal design* problem is the SDP

$$\begin{array}{ll} \text{maximize}_{w,t} & t \\ \text{subject to} & M(w) - tI \succeq 0 \\ & w \ge 0, \quad \mathbf{1}^T w = 1. \end{array}$$

(The "E" stands for eigenvalue. Strictly speaking, Nw_i should be an integer; treating w as continuous is itself a relaxation, which is accurate when N is large.)

Figure 15 shows what the SDP chooses. The E-optimal design spends 58% of the budget on the noisy sensor that measures the neglected direction, and it drops three of the five redundant sensors entirely; the weights are approximately $(0.209, 0, 0, 0, 0.209, 0.581)$. The smallest eigenvalue of $M(w)$ rises from 0.166 (uniform) to 0.314, so the standard deviation in the worst direction, $1/\sqrt{\lambda_{\min}}$, shrinks from about 2.46 to 1.78.

Exercise 7.3. Suppose there are only two candidate sensors, $v_1 = e_1$ and $v_2 = \frac{1}{2}e_2$ in $\mathbb{R}^2$. Compute $M(w)$, find the E-optimal weights by hand, and give the optimal value of $\lambda_{\min}(M(w))$.

7.4 Lifting products: MaxCut

We now meet an SDP that is *not* an exact reformulation, but a relaxation of a hard problem.

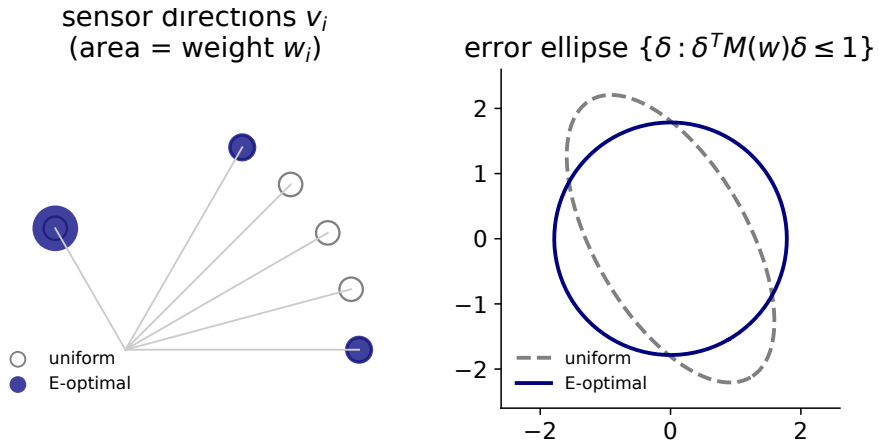


Figure 15: Left: the six sensors, with marker area proportional to the weight w_i, for the uniform and the E-optimal design. Right: the corresponding error ellipses.

Graphs and cuts. A *graph* $G = (V, E)$ has vertices $V = \{1, \dots, n\}$ and a set E of undirected edges $\{i, j\}$. Each edge has a weight $w_{ij} \ge 0$; set $w_{ij} = 0$ when $\{i, j\} \notin E$, and collect the weights in a symmetric matrix W with zero diagonal. A *cut* splits the vertices into two groups. We encode it by assigning each vertex a sign $x_i \in \{-1, 1\}$. Then

$$\frac{1 - x_i x_j}{2} = \begin{cases} 1 & \text{if } x_i \neq x_j \text{ (edge } \{i, j\} \text{ crosses the cut)}, \\ 0 & \text{if } x_i = x_j. \end{cases}$$

The *MaxCut* problem asks for the cut of largest total weight:

$$\underset{x \in \{-1,1\}^n}{\text{maximize}} \sum_{\{i,j\} \in E} w_{ij} \frac{1 - x_i x_j}{2},$$

where each undirected edge is counted once. MaxCut is NP-hard, so we do not expect an efficient exact algorithm.

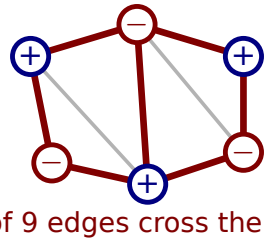


Figure 16: A cut of a small graph. Vertices are labeled by their sign; seven of the nine edges cross the cut.

Lifting. The objective is quadratic in x. Can we make it linear by introducing a matrix? Let $X = xx^T$. Then $X_{ij} = x_i x_j$, so the objective becomes $\sum_{\{i,j\} \in E} w_{ij}(1 - X_{ij})/2$, which is linear in

X. What constraints does X satisfy? A matrix X equals xx^T for some $x \in \{-1,1\}^n$ if and only if

$$X \succeq 0, \qquad X_{ii} = 1 \;\; (i = 1, \ldots, n), \qquad \mathrm{Rank}(X) = 1.$$

(If X has these properties, then $X = xx^T$ for some x by the rank fact in Section 2, and $x_i^2 = X_{ii} = 1$ forces $x_i = \pm 1$.) So MaxCut is *exactly* the lifted problem

$$\begin{aligned} &\text{maximize}_X \quad \sum_{\{i,j\} \in E} w_{ij} \frac{1 - X_{ij}}{2} \\ &\text{subject to} \quad X \succeq 0, \quad X_{ii} = 1, \quad \mathrm{Rank}(X) = 1. \end{aligned}$$

7.5 Where does the relaxation happen?

The objective is linear, and the constraints $X \succeq 0$ and $X_{ii} = 1$ are convex. The only nonconvex constraint is $\mathrm{Rank}(X) = 1$. If we drop it, we obtain the SDP

$$\begin{aligned} &\text{maximize}_X \quad \sum_{\{i,j\} \in E} w_{ij} \frac{1 - X_{ij}}{2} \\ &\text{subject to} \quad X \succeq 0, \quad X_{ii} = 1, \;\; i = 1, \ldots, n. \end{aligned}$$

Every cut is still feasible (as $X = xx^T$), so this is a *relaxation*. Since we maximize over a larger set, its optimal value is an *upper bound* on the best cut. Unlike the epigraph trick, this step may change the optimal value. The same lift works for any quadratic objective over $x \in \{-1,1\}^n$; it is called the *Shor relaxation*.

Exercise 7.4. Use Exercise 2.5 to show that the MaxCut SDP value is at most the total edge weight $\sum_{\{i,j\} \in E} w_{ij}$. Conclude that the relaxation is tight for the 4-cycle (four vertices joined in a ring by four edges of weight 1).

7.6 A triangle exposes the gap

Take three vertices with all three edges present, each of weight 1. A cut can cross at most two edges: with only two signs available, two of the three vertices share a sign. So the best cut has value 2. Now consider

$$X = \begin{bmatrix} 1 & -1/2 & -1/2 \\ -1/2 & 1 & -1/2 \\ -1/2 & -1/2 & 1 \end{bmatrix} = \frac{3}{2} I - \frac{1}{2} \mathbf{1}\mathbf{1}^T.$$

The eigenvalues of $\mathbf{1}\mathbf{1}^T$ are $3, 0, 0$, so the eigenvalues of X are $\frac{3}{2} - \frac{3}{2} = 0$ and $\frac{3}{2}, \frac{3}{2}$. Hence $X \succeq 0$, its diagonal is 1, and it is feasible for the SDP. Its objective value is

$$3 \cdot \frac{1 - (-1/2)}{2} = \frac{9}{4} > 2.$$

So the SDP bound $9/4$ (we will prove it is optimal in Section 8) exceeds the best cut. What prevents this X from representing a cut? It has rank two. Factoring $X = VV^T$, the rows v_1, v_2, v_3 are unit vectors in a plane, 120° apart (Figure 17), since $v_i^T v_j = \cos 120^\circ = -\frac{1}{2}$. No assignment of signs can make all three pairs opposite, but three vectors in a plane can all be "partly opposite."

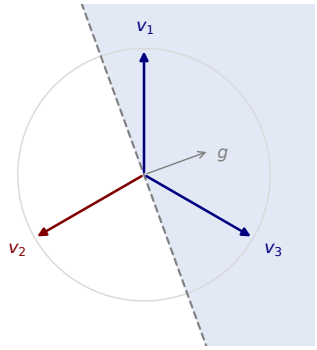


Figure 17: The factor V of the optimal triangle SDP solution: three unit vectors 120° apart. A random line through the origin (dashed, with normal g) splits them into two groups.

7.7 Recovering a decision: randomized rounding

A relaxation gives a bound, but we usually also want a decision. Factor a feasible SDP matrix as $X = VV^T$; since $X_{ii} = 1$, its rows v_i are unit vectors.

- If X has rank one, the v_i are all $\pm$ the same vector, and we recover the signs x_i exactly.
- Otherwise, choose a random direction g (for example, Gaussian) and set $x_i = \mathbf{sign}(v_i^T g)$, breaking ties arbitrarily. Geometrically, a random hyperplane through the origin splits the vectors into two groups. Evaluate the resulting cut.

For any cut we produce,

$$\text{value of the rounded cut} \;\leq\; \text{best cut value} \;\leq\; \text{SDP optimal value.}$$

So we have both a decision and a measure of how far from optimal it might be.

Theorem 7.5 (Goemans–Williamson, 1995)**.** *Let $w \geq 0$, let $X^\star$ be optimal for the MaxCut SDP, and round with a Gaussian g. Then*

$$\mathbf{E}[\textit{rounded cut}] \;\geq\; 0.878 \cdot \textit{SDP value} \;\geq\; 0.878 \cdot \textit{best cut.}$$

Why 0.878? Two unit vectors at angle θ are separated by a random hyperplane with probability θ/π (the hyperplane's trace in their plane is a random line, and it separates them when it falls in the angle between them). The edge contributes $(1 - \cos\theta)/2$ to the SDP objective. A calculus exercise shows that $\theta/\pi \geq 0.878 \cdot (1-\cos\theta)/2$ for all $\theta \in [0, \pi]$. Summing over edges with weights $w_{ij} \geq 0$ gives the theorem.

Gotcha 7.6 (A feasible matrix is not a bound). The SDP optimal value bounds the best cut. A merely feasible X does not: its value can be anywhere below the SDP optimum. To use the SDP as a certified upper bound, we need the optimal value itself, or a certified upper bound from the dual, which is the subject of Section 8.

How good is rounding in practice? Figure 18 shows a random graph with 20 vertices and 67 edges, small enough to check all 2^{19} cuts by brute force. The best cut is 48 and the SDP bound is 49.6, so the relaxation is within 4%. Over 5000 random roundings, the mean cut is 45.6, about 0.92 times the bound, and 7% of roundings find a best cut. A relaxation is useful when it provides a bound, a good decision, or both.

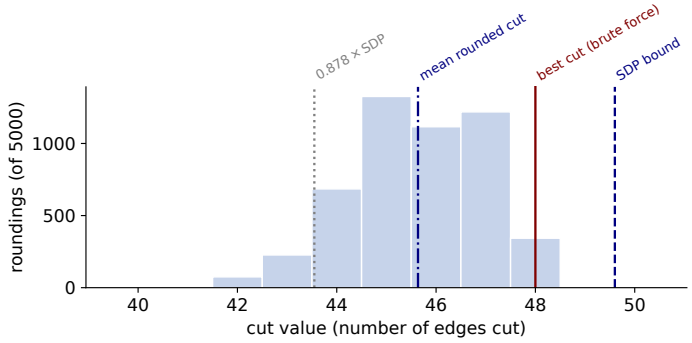


Figure 18: Histogram of cut values from 5000 Goemans–Williamson roundings, with the 0.878 guarantee, the mean rounded cut, the best cut (brute force), and the SDP bound.

Exercise 7.7. For the triangle, round the vectors of Figure 17 with a random hyperplane. How many edges does the rounded cut cross? Compare with the guarantee of Goemans and Williamson.

8 Conic duality

Conic duality is not a new theory. It is the Lagrangian recipe from the convex duality lecture, applied to a cone constraint. The only new ingredient is the right sign condition for the multipliers, and we will derive it from one principle: *the Lagrangian is built to be a lower bound.*

8.1 Recall: the Lagrangian is built to be a lower bound

Consider $\min\{f_0(x) : f_i(x) \le 0,\ i = 1, \dots, m,\ x \in \mathcal{X}\}$ with optimal value $p^\star$. For $\lambda \ge 0$, the Lagrangian

$$\mathcal{L}(x, \lambda) = f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x)$$

satisfies $\mathcal{L}(x, \lambda) \le f_0(x)$ for every feasible x, since each added term $\lambda_i f_i(x)$ is a nonnegative number times a nonpositive number. Hence the dual function $g(\lambda) = \inf_{x \in \mathcal{X}} \mathcal{L}(x, \lambda)$ satisfies $g(\lambda) \le p^\star$, and the dual problem looks for the best such bound. The sign condition $\lambda \ge 0$ is *exactly* what makes each added term nonpositive on feasible points.

Now the constraint is $s = b - Ax \in K$, and we add the term $-\langle \lambda, s \rangle$ to the objective. Which multipliers λ make $-\langle \lambda, s \rangle \le 0$ for *every* $s \in K$? That question defines the dual cone.

8.2 Dual cones

Definition 8.1 (Dual cone)**.** The *dual cone* of a cone K is

$$K^* = \{\lambda : \langle \lambda, s\rangle \geq 0 \text{ for all } s \in K\}.$$

A cone is *self-dual* if $K^* = K$.

So K^* is precisely the set of multipliers that keep the Lagrangian a lower bound. Geometrically, $\lambda \in K^*$ means that λ makes an angle of at most 90° with every vector in K (Figure 19). The dual cone is always a closed convex cone, since it is an intersection of halfspaces through the origin.

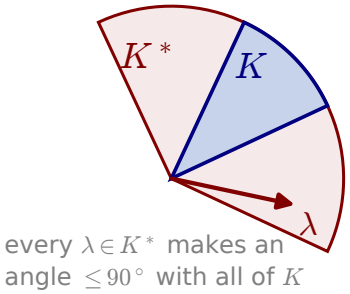


Figure 19: A cone K and its dual cone K^* in the plane. Every $\lambda \in K^*$ makes an angle of at most 90° with every vector in K. A narrow cone has a wide dual.

Two dual cones recover familiar rules.

- $(\mathbb{R}^m_+)^* = \mathbb{R}^m_+$. If $\lambda \geq 0$ then $\lambda^T s \geq 0$ for all $s \geq 0$; conversely, testing against $s = e_i$ gives $\lambda_i \geq 0$. This is the usual rule $\lambda \geq 0$ for inequality constraints.
- $\{0\}^* = \mathbb{R}^m$: every λ satisfies $\langle \lambda, 0\rangle \geq 0$. This is the usual rule that multipliers for equality constraints are free.

For a product cone, the dual is the product of the duals, $(K_1 \times K_2)^* = K_1^* \times K_2^*$, so we attach a multiplier to each block separately.

Proposition 8.2 (Self-dual cones)**.** *The second-order cone and the PSD cone are self-dual:*

$$(\mathcal{Q}^{m+1})^* = \mathcal{Q}^{m+1}, \qquad (\mathbf{S}^m_+)^* = \mathbf{S}^m_+,$$

where the PSD cone uses the trace inner product $\langle X, Y\rangle = \mathbf{tr}(XY)$.

Proof. Second-order cone. First, $\mathcal{Q} \subseteq \mathcal{Q}^*$. Let (z, t) and (u, τ) lie in $\mathcal{Q}$. By Cauchy–Schwarz,

$$z^T u + t\tau \geq -\|z\|_2\|u\|_2 + t\tau \geq 0,$$

since $t \geq \|z\|_2 \geq 0$ and $\tau \geq \|u\|_2 \geq 0$. Second, $\mathcal{Q}^* \subseteq \mathcal{Q}$. Let $(u, \tau) \in \mathcal{Q}^*$, and test it against the point $(-u, \|u\|_2) \in \mathcal{Q}$:

$$0 \leq -u^T u + \tau\|u\|_2 = \|u\|_2(\tau - \|u\|_2).$$

If $u \neq 0$ this gives $\tau \geq \|u\|_2$. If $u = 0$, test against $(0,1) \in \mathcal{Q}$ to get $\tau \geq 0$. Either way $(u,\tau) \in \mathcal{Q}$.

PSD cone. First, $\mathbf{S}_+ \subseteq \mathbf{S}_+^*$. If $X, Y \succeq 0$, write $X = \sum_i \lambda_i(X) q_i q_i^T$ with $\lambda_i(X) \geq 0$; then $\langle X, Y\rangle = \sum_i \lambda_i(X) q_i^T Y q_i \geq 0$. Second, $\mathbf{S}_+^* \subseteq \mathbf{S}_+$. If $Y \in \mathbf{S}_+^*$, then for every z the matrix zz^T is PSD, so $0 \leq \langle Y, zz^T\rangle = z^T Y z$. Hence $Y \succeq 0$. □

Exercise 8.3. Show that $(K_1 \times K_2)^* = K_1^* \times K_2^*$ for any cones K_1 and K_2. (Hint: for one inclusion, test against points of the form $(s_1, 0)$ and $(0, s_2)$.)

8.3 The conic dual: the same recipe

Consider the conic program

$$\mathcal{P}: \quad \begin{array}{ll} \text{minimize}_x & c^T x \\ \text{subject to} & b - Ax \in K \end{array}$$

with optimal value $p^\star$. Follow the recipe in three steps.

1. *Attach a multiplier* $\lambda \in K^*$ to the constraint $b - Ax \in K$ (block by block, for a product cone).
2. *Check the lower bound.* Define $\mathcal{L}(x,\lambda) = c^T x - \langle \lambda, b - Ax\rangle$. For feasible x, the slack $b - Ax$ lies in K and $\lambda \in K^*$, so $\langle \lambda, b - Ax\rangle \geq 0$ and $\mathcal{L}(x,\lambda) \leq c^T x$.
3. *Minimize over* x. Rearranging, $\mathcal{L}(x,\lambda) = -b^T\lambda + (c + A^T\lambda)^T x$. This is affine in x, and x is unrestricted. If $c + A^T\lambda \neq 0$, moving x in the direction $-(c + A^T\lambda)$ drives $\mathcal{L}$ to $-\infty$. So

$$g(\lambda) = \inf_x \mathcal{L}(x,\lambda) = \begin{cases} -b^T\lambda & \text{if } c + A^T\lambda = 0, \\ -\infty & \text{otherwise.} \end{cases}$$

The best lower bound gives the *conic dual problem*

$$\mathcal{D}: \quad \begin{array}{ll} \text{maximize}_\lambda & -b^T\lambda \\ \text{subject to} & c + A^T\lambda = 0 \\ & \lambda \in K^*, \end{array}$$

with optimal value $d^\star$.

Theorem 8.4 (Weak conic duality)**.** *If x is feasible for $\mathcal{P}$ and λ is feasible for $\mathcal{D}$, then*

$$c^T x - (-b^T\lambda) = \langle \lambda, b - Ax\rangle \geq 0.$$

Consequently $d^\star \leq p^\star$.

Proof. Using $c = -A^T\lambda$, we get $c^T x + b^T\lambda = -\lambda^T Ax + b^T\lambda = \langle \lambda, b - Ax\rangle$, which is nonnegative because $b - Ax \in K$ and $\lambda \in K^*$. □

Two remarks extend the recipe.

- With matrix constraints, replace A^T by the *adjoint* of the linear map $\mathcal{A}$: the linear map $\mathcal{A}^*$ with $\langle \lambda, \mathcal{A}(x)\rangle = \langle \mathcal{A}^*(\lambda), x\rangle$ for all x, λ.

- For a *maximization* problem, build an *upper* bound instead: choose the signs so that each added term is nonnegative on feasible points, then maximize over the primal variables.

Exercise 8.5. Use the recipe to derive the dual of the LP $\min\{c^T x : Fx \le g\}$. Then derive the dual of the standard-form LP $\min\{c^T x : Fx = g,\ x \ge 0\}$. (Hint: write the constraints as $b - Ax \in \{0\}^m \times \mathbb{R}^n_+$.)

Exercise 8.6. Let $c \in \mathbb{R}^n$ be nonzero. Write $\min\{c^T x : \|x\|_2 \le 1\}$ as a conic program, derive its dual, and check that $p^\star = d^\star = -\|c\|_2$.

8.4 Example: the dual of the MaxCut SDP

First, write the MaxCut SDP objective as $\langle C, X\rangle$. The *graph Laplacian* is

$$L = \mathbf{Diag}(W\mathbf{1}) - W,$$

where $\mathbf{Diag}(y)$ denotes the diagonal matrix with diagonal y; so L_{ii} is the total weight of edges at vertex i, and $L_{ij} = -w_{ij}$ for $i \ne j$. For any symmetric X,

$$\langle L, X\rangle = \sum_{\{i,j\}\in E} w_{ij}(X_{ii} + X_{jj} - 2X_{ij}),$$

because each edge $\{i,j\}$ contributes w_{ij} to L_{ii} and L_{jj} and $-w_{ij}$ to L_{ij} and L_{ji}. When $X_{ii} = 1$ the right-hand side is $\sum_{\{i,j\}\in E} 2w_{ij}(1 - X_{ij})$. Hence with $C = \frac{1}{4}L$ the MaxCut SDP reads

$$\begin{array}{ll} \text{maximize}_X & \langle C, X\rangle \\ \text{subject to} & X_{ii} = 1, \quad i = 1, \dots, n \\ & X \succeq 0. \end{array}$$

Since we maximize, we build an *upper* bound. Attach a free multiplier λ_i to each equality $X_{ii} = 1$ (the dual of the zero cone is everything) and a multiplier $Z \succeq 0$ to $X \succeq 0$ (the PSD cone is self-dual).[1] Define

$$\mathcal{L}(X, \lambda, Z) = \langle C, X\rangle + \sum_{i=1}^{n} \lambda_i(1 - X_{ii}) + \langle Z, X\rangle.$$

For feasible X, the middle sum is zero and $\langle Z, X\rangle \ge 0$ by Proposition 8.2, so $\mathcal{L}(X, \lambda, Z) \ge \langle C, X\rangle$. Next maximize over all symmetric X. Using $\sum_i \lambda_i X_{ii} = \langle \mathbf{Diag}(\lambda), X\rangle$,

$$\mathcal{L}(X, \lambda, Z) = \mathbf{1}^T\lambda + \langle C - \mathbf{Diag}(\lambda) + Z,\ X\rangle.$$

This is bounded above in X only if $C - \mathbf{Diag}(\lambda) + Z = 0$, and then it equals $\mathbf{1}^T\lambda$. Eliminating $Z = \mathbf{Diag}(\lambda) - C \succeq 0$ gives the dual SDP

$$\begin{array}{ll} \text{minimize}_\lambda & \mathbf{1}^T\lambda \\ \text{subject to} & \mathbf{Diag}(\lambda) - C \succeq 0. \end{array}$$

Every feasible λ certifies that *no* cut, and indeed no feasible X, has value above $\mathbf{1}^T\lambda$.

[1] The slides call the equality multiplier y.

8.5 Certify the triangle

For the triangle, $W = \mathbf{1}\mathbf{1}^T - I$ and $W\mathbf{1} = 2\mathbf{1}$, so $L = 2I - W = 3I - \mathbf{1}\mathbf{1}^T$ and $C = \frac{1}{4}(3I - \mathbf{1}\mathbf{1}^T)$. We want λ with $\mathbf{Diag}(\lambda) - C \succeq 0$ and $\mathbf{1}^T\lambda = 9/4$. By symmetry, try $\lambda = \frac{3}{4}\mathbf{1}$:

$$\mathbf{Diag}(\lambda) - C = \tfrac{3}{4}I - \tfrac{3}{4}I + \tfrac{1}{4}\mathbf{1}\mathbf{1}^T = \tfrac{1}{4}\mathbf{1}\mathbf{1}^T \succeq 0, \qquad \mathbf{1}^T\lambda = \tfrac{9}{4}.$$

Now put the pieces together.

- By weak duality, no feasible X, and so no cut, has value above $9/4$.
- The matrix $X = \frac{3}{2}I - \frac{1}{2}\mathbf{1}\mathbf{1}^T$ from before attains $9/4$. The primal and dual values agree (zero gap), so both are optimal.
- Complementary slackness holds: with $Z = \frac{1}{4}\mathbf{1}\mathbf{1}^T$, $\langle Z, X\rangle = \frac{1}{4}\mathbf{1}^T X\mathbf{1} = 0$, because $X\mathbf{1} = \frac{3}{2}\mathbf{1} - \frac{3}{2}\mathbf{1} = 0$.

This is a 3×3 certificate you can check by hand, without trusting any solver.

8.6 Example: the dual of square-root Lasso

To dualize the square-root Lasso, it is convenient to keep the residual as a variable:

$$\begin{array}{ll} \text{minimize}_{x,r} & \|r\|_2 + \gamma\|x\|_1 \\ \text{subject to} & r = Ax - b. \end{array}$$

Try deriving the dual yourself first: attach a multiplier to $r = Ax - b$ (what sign does it need?), minimize the Lagrangian over r and over x separately, and ask which multipliers give a finite lower bound.

We need one more definition.

Definition 8.7 (Dual norm)**.** The *dual norm* of a norm $\|\cdot\|$ is

$$\|\lambda\|_* = \sup_{\|r\| \le 1} \lambda^T r.$$

By (1) and (2), the dual of $\|\cdot\|_2$ is $\|\cdot\|_2$ and the dual of $\|\cdot\|_1$ is $\|\cdot\|_\infty$.

Proposition 8.8. *For any norm and any λ,*

$$\inf_r \left(\|r\| - \lambda^T r\right) = \begin{cases} 0 & \text{if } \|\lambda\|_* \le 1, \\ -\infty & \text{otherwise.} \end{cases}$$

Proof. If $\|\lambda\|_* \le 1$, then $\lambda^T r \le \|\lambda\|_*\|r\| \le \|r\|$ for every r (scale the definition), so the expression is nonnegative, and $r = 0$ gives 0. If $\|\lambda\|_* > 1$, pick r_0 with $\|r_0\| \le 1$ and $\lambda^T r_0 > 1$. Along $r = \alpha r_0$ with $\alpha > 0$, the expression is at most $\alpha(1 - \lambda^T r_0)$, which tends to $-\infty$. □

Now attach a multiplier $\lambda \in \mathbb{R}^m$ to the equality $r = Ax - b$. It is free, because an equality constraint adds the term $\lambda^T(Ax - b - r) = 0$ on feasible points whatever the sign of λ.[2] Grouping terms,

$$\mathcal{L}(x, r, \lambda) = \big(\|r\|_2 - \lambda^T r\big) + \big(\gamma\|x\|_1 + \lambda^T A x\big) - b^T\lambda.$$

The variables separate, so we minimize each group on its own using Proposition 8.8:

- $\inf_r \big(\|r\|_2 - \lambda^T r\big) = 0$ if $\|\lambda\|_2 \le 1$, and $-\infty$ otherwise;
- $\inf_x \big(\gamma\|x\|_1 + (A^T\lambda)^T x\big) = 0$ if $\|A^T\lambda\|_\infty \le \gamma$, and $-\infty$ otherwise (apply the proposition to $\gamma\|x\|_1$ and the vector $-A^T\lambda/\gamma$; the sign does not matter for a norm bound).

The dual problem is therefore

$$\begin{array}{ll} \text{maximize}_\lambda & -b^T\lambda \\ \text{subject to} & \|\lambda\|_2 \le 1, \quad \|A^T\lambda\|_\infty \le \gamma. \end{array}$$

Dual norms appear naturally: ℓ_2 pairs with ℓ_2, and ℓ_1 pairs with ℓ_∞. They are tied to dual cones as well: the dual of the norm cone $\{(r, t) : \|r\| \le t\}$ is $\{(\lambda, \tau) : \|\lambda\|_* \le \tau\}$. For the Euclidean norm, this is the self-duality of $\mathcal{Q}$.

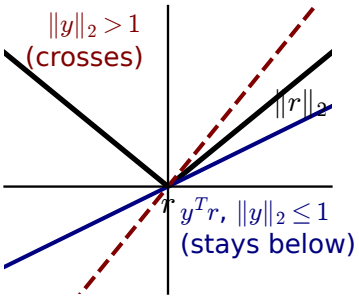


Figure 20: The condition $\|\lambda\|_2 \le 1$ in one dimension: the line λr stays below $|r|$ exactly when its slope has magnitude at most one. A steeper line crosses $|r|$, and $\|r\| - \lambda r$ is unbounded below. (The figure writes y for λ.)

Exercise 8.9. Take $m = n = 1$, $A = 1$, and $b = 2$, so the primal problem is $\min_x |x - 2| + \gamma|x|$. Solve the primal and the dual by hand for $\gamma = \frac{1}{2}$ and for $\gamma = 2$, and check that the optimal values agree. What does the solution look like when $\gamma \ge 1$?

8.7 The gap is a certificate

Weak duality turns any primal–dual feasible pair into a computable error bound. For primal feasible x and dual feasible λ, let $s = b - Ax \in K$. Then

$$0 \le c^T x - p^\star \le c^T x - (-b^T\lambda) = \langle \lambda, s \rangle.$$

The quantity $\langle \lambda, s \rangle$ is the *duality gap* of the pair. It bounds how suboptimal x is, and it uses only quantities we can compute. If the gap is zero, then x and λ are both optimal. The condition $\langle \lambda, s \rangle = 0$ is *complementary slackness* for conic programs.

[2] The slides call this multiplier y.

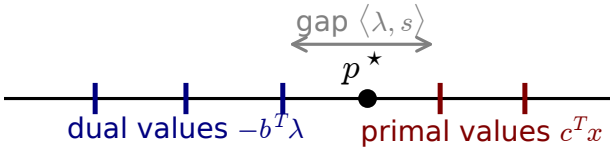


Figure 21: Dual values lie below $p^\star$ and primal values lie above it. The gap between a dual value and a primal value brackets $p^\star$.

When can we expect a zero gap? As for general convex problems, we need a constraint qualification. The *interior* of the standard cones is easy to describe:

$$\mathbf{int}\,\mathbb{R}^m_+ = \{s : s > 0\}, \qquad \mathbf{int}\,\mathcal{Q}^{m+1} = \{(z,t) : \|z\|_2 < t\}, \qquad \mathbf{int}\,\mathbf{S}^m_+ = \{X : X \succ 0\}.$$

The zero cone has no interior, but its *relative interior* is $\{0\}$ itself. For a product cone, the relative interior is the product of the relative interiors.

Theorem 8.10 (Slater's condition for conic programs)**.** *Suppose $p^\star$ is finite and some x satisfies $b - Ax \in \mathbf{relint}\,K$. Then $p^\star = d^\star$, and the dual optimum is attained.*

In practice: satisfy the explicit equality constraints exactly, and satisfy every other cone block strictly (positive slacks, $\|z\|_2 < t$, positive definite matrices).

Checking a computed solution. Solvers return approximate solutions. To check one, compute in this order:

1. the primal residual: is $b - Ax$ in K, up to a small tolerance?
2. the dual residual: is $c + A^T\lambda \approx 0$, and is $\lambda \in K^*$?
3. *then* the gap $\langle \lambda, b - Ax \rangle$.

The order matters: the gap bounds suboptimality only for primal and dual *feasible* points. A tiny reported gap says nothing about an infeasible decision.

Watching the certificate form. Figure 22 shows the square-root Lasso with $(m, n) = (300, 150)$. At every Clarabel iterate k we compute an upper bound, the original objective at x_k (every x is feasible for this unconstrained problem), and a lower bound, $-b^T\lambda_k$, after scaling λ_k so that $\|\lambda\|_2 \le 1$ and $\|A^T\lambda\|_\infty \le \gamma$. The certified gap falls from 1.8 to 10^{-9} in 12 iterations.

8.8 Certificates of infeasibility

Duality can also prove that *no* feasible point exists.

Proposition 8.11 (Conic Farkas certificate)**.** *If $\lambda \in K^*$, $A^T\lambda = 0$, and $b^T\lambda < 0$, then no x satisfies $b - Ax \in K$.*

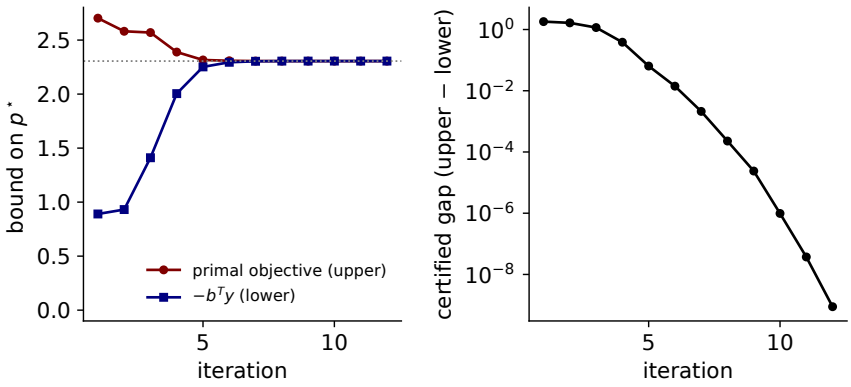


Figure 22: Left: certified upper and lower bounds on $p^\star$ at each Clarabel iterate for square-root Lasso. Right: the certified gap on a log scale.

Proof. If some x had $b - Ax \in K$, then, since $\lambda \in K^*$,

$$0 \le \langle \lambda, b - Ax \rangle = b^T\lambda - (A^T\lambda)^T x = b^T\lambda < 0,$$

a contradiction. This is the Lagrangian bound once more. □

For LPs, Farkas' lemma says that such a certificate always exists when the problem is infeasible. For general cones the converse needs mild extra conditions, but conic solvers return a certificate of this kind when they declare a problem infeasible.

Example 8.12 (An infeasible robust production plan)**.** Return to robust production with $\rho = 2$, and add a minimum output requirement $x_1 + x_2 \ge 0.9$. Here $\rho P^T x = (x_1, 0.2x_2)$, and the robust constraint reads $\|(x_1, 0.2x_2)\|_2 \le 1 - x_1 - x_2$. Let $u = (0.15, 0.75)$, so $\|u\|_2 = \sqrt{0.585} \approx 0.765$. For any x satisfying the robust constraint, Cauchy–Schwarz gives

$$0.15(x_1 + x_2) = u^T \begin{pmatrix} x_1 \\ 0.2x_2 \end{pmatrix} \le 0.765 \left\| \begin{pmatrix} x_1 \\ 0.2x_2 \end{pmatrix} \right\|_2 \le 0.85(1 - x_1 - x_2),$$

where the last step uses the robust constraint and $1 - x_1 - x_2 \ge 0$. Rearranging, $x_1 + x_2 \le 0.85 < 0.9$. So the problem is infeasible, and the chain of inequalities above is a proof you can check.

In conic form, the certificate puts multiplier 1 on the demand constraint $x_1 + x_2 - 0.9 \ge 0$, the multiplier $(-0.15, -0.75, 0.85) \in \mathcal{Q}^3$ on the block $(x_1, 0.2x_2, 1 - x_1 - x_2) \in \mathcal{Q}^3$, and 0 on $x \ge 0$ (Exercise 8.13). Clarabel reports `PrimalInfeasible` after 8 iterations and returns a certificate like this one. IPOPT, given the squared formulation, reports "infeasible problem detected" after 16 iterations, which is a heuristic conclusion, not a proof.

Exercise 8.13. Write the infeasible robust production problem as $b - Ax \in K$ with $K = \mathbb{R}_+ \times \mathcal{Q}^3 \times \mathbb{R}^2_+$. Verify that the multiplier $\lambda = (1; -0.15, -0.75, 0.85; 0, 0)$ satisfies $\lambda \in K^*$, $A^T\lambda = 0$, and $b^T\lambda < 0$.

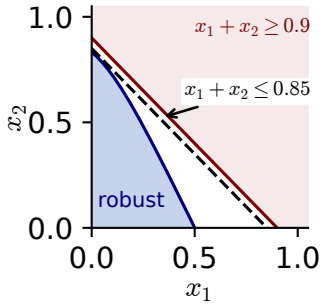


Figure 23: The robust feasible set (blue) and the demand region $x_1 + x_2 \geq 0.9$ (red). The certificate proves that the line $x_1 + x_2 = 0.85$ separates them.

8.9 Midterm practice: robust counterpart by conic duality

Earlier, LP duality gave the robust counterpart for a polyhedral uncertainty set. The next exercise repeats that argument for a ball, using conic duality in place of Cauchy–Schwarz. Try it on your own before reading the solution.

Exercise 8.14. Fix $v \in \mathbb{R}^k$ and $\rho > 0$.

(a) Write $\sup\{v^T u : \|u\|_2 \leq \rho\}$ as a conic program in u.

(b) Build its Lagrangian so that it is an *upper* bound, and derive the dual.

(c) Conclude that $\bar{a}^T x + \rho\|P^T x\|_2 \leq b$ holds if and only if there exists τ with

$$\bar{a}^T x + \rho\tau \leq b, \qquad \|P^T x\|_2 \leq \tau.$$

(d) Where did you use strong duality? Which constraint qualification holds?

9 Solutions to exercises

Try each exercise before reading its solution. Many have short answers; the reasoning is what matters.

Exercise 2.5. Taking $z = e_i$ gives $X_{ii} = e_i^T X e_i \geq 0$. Next take $z = \alpha e_i + \beta e_j$:

$$0 \leq z^T X z = \alpha^2 X_{ii} + 2\alpha\beta X_{ij} + \beta^2 X_{jj} = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}^T \begin{bmatrix} X_{ii} & X_{ij} \\ X_{ij} & X_{jj}\end{bmatrix} \begin{pmatrix}\alpha\\ \beta\end{pmatrix}.$$

Since this holds for all (α, β), the 2×2 matrix is PSD. Its eigenvalues are nonnegative, so their product, the determinant $X_{ii}X_{jj} - X_{ij}^2$, is nonnegative. With unit diagonal this reads $X_{ij}^2 \leq 1$.

Exercise 2.6. The eigenvalues are 3 and 1, with eigenvectors $(1,1)/\sqrt{2}$ and $(1,-1)/\sqrt{2}$, respectively. Both eigenvalues are positive, so $X \succ 0$ and in particular $X \succeq 0$. By Proposition 2.3, $z^T X z$ over unit vectors is maximized by $z = \pm(1,1)/\sqrt{2}$, with value $\lambda_{\max}(X) = 3$.

Exercise 3.6. With $\rho = 1$, $\rho P^T x = (0.5x_1, 0.1x_2)$. At $x = (2/3, 0)$ the robust resource use is $\frac{2}{3} + \|(\frac{1}{3}, 0)\|_2 = \frac{2}{3} + \frac{1}{3} = 1$, so the constraint holds with equality. The profit is $3 \cdot \frac{2}{3} = 2$. (A solver confirms that this plan is optimal for $\rho = 1$.) This exceeds the $\rho = 2$ profit of 1.7534, as it must: the $\rho = 2$ feasible set is contained in the $\rho = 1$ feasible set, so the $\rho = 1$ problem maximizes over a larger set.

Exercise 3.8. The violation probability is $1-\Phi(\rho)$. Setting $1-\Phi(\rho) = 0.01$ gives $\rho = \Phi^{-1}(0.99) \approx 2.326$.

Exercise 3.9. By (2), the worst extra use is $2\|P^T x\|_1 = 2(0.5|x_1|+0.1|x_2|)$, which equals $x_1+0.2x_2$ for $x \geq 0$. The robust counterpart is $x_1 + x_2 + x_1 + 0.2x_2 \leq 1$, that is, $2x_1 + 1.2x_2 \leq 1$. With one resource constraint, put everything into the product with the best profit per unit of resource: product 1 earns $3/2 = 1.5$, and product 2 earns $2/1.2 \approx 1.667$. So $x^\star = (0, 5/6)$ with profit $5/3 \approx 1.667$. This is below the ball model's 1.7534: the box contains the ball, so it is more conservative.

Exercise 5.2. The SOC is nonempty (it contains 0). Let $(z_1, t_1), (z_2, t_2) \in \mathcal{Q}^{m+1}$ and $\alpha, \beta \geq 0$. By the triangle inequality and homogeneity of the norm,

$$\|\alpha z_1 + \beta z_2\|_2 \leq \alpha\|z_1\|_2 + \beta\|z_2\|_2 \leq \alpha t_1 + \beta t_2,$$

so $\alpha(z_1, t_1)+\beta(z_2, t_2) \in \mathcal{Q}^{m+1}$. For the PSD cone, let $X, Y \succeq 0$ and $\alpha, \beta \geq 0$. The matrix $\alpha X + \beta Y$ is symmetric, and for every z, $z^T(\alpha X + \beta Y)z = \alpha z^T X z + \beta z^T Y z \geq 0$.

Exercise 5.9. Introduce t with $\|Ax - b\|_\infty \leq t$, which holds if and only if $-t \leq a_i^T x - b_i \leq t$ for every row i. The conic form is

$$\begin{array}{ll} \text{minimize}_{x,t} & t \\ \text{subject to} & t\mathbf{1} - (Ax - b) \geq 0, \quad t\mathbf{1} + (Ax - b) \geq 0, \end{array}$$

with cone $\mathbb{R}^m_+ \times \mathbb{R}^m_+$, so this is an LP. It is exact by Proposition 5.6: $(x, \|Ax - b\|_\infty)$ is feasible with the same value, and every feasible (x, t) has $t \geq \|Ax - b\|_\infty$.

Exercise 5.10. The constraint is $(Fx+g,\ h^T x+d) \in \mathcal{Q}^{k+1}$. For $\|x\|_2 \leq 1$, we need $b-Ax = (x, 1)$, so

$$A = \begin{bmatrix} -I_n \\ 0^T \end{bmatrix} \in \mathbb{R}^{(n+1)\times n}, \qquad b = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \in \mathbb{R}^{n+1}.$$

Exercise 5.11. Suppose $\|(2z, t-1)\|_2 \leq t+1$. Then $t+1 \geq 0$, and squaring both sides gives $4\|z\|_2^2 + (t-1)^2 \leq (t+1)^2$, that is, $4\|z\|_2^2 \leq 4t$, so $\|z\|_2^2 \leq t$. Conversely, if $\|z\|_2^2 \leq t$, then $t \geq 0$, so $t+1 > 0$; the same algebra run backwards gives the squared inequality, and taking square roots of two nonnegative numbers gives the SOC inequality. For $x^T H x = \|Rx\|_2^2 \leq t$, take $z = Rx$:

$$\big((2Rx,\ t-1),\ t+1\big) \in \mathcal{Q}^{k+2},$$

where k is the number of rows of R. All entries are affine in (x, t).

Exercise 6.3.

(a) DCP-convex: exp is convex and nondecreasing, and $\|y\|_2$ is convex.

(b) DCP-convex: $\sqrt{x}$ is a concave library function, and its negative is convex.

(c) Not DCP: $\sqrt{\cdot}$ is concave, so no rule applies to $\sqrt{\text{convex}}$. The function is not convex either: along $y = (s, 0)$ with $s > 0$ it equals $\sqrt{s}$, which is strictly concave.

(d) Not DCP as written, for the same reason as $\sqrt{1 + x^2}$. But it is convex: $\sqrt{x^2 + 4} = \|(x, 2)\|_2$, a norm of an affine expression, which is DCP.

Exercise 7.2. The trace of $X(x)$ is 0 and its determinant is $-x^2 - 1$, so its eigenvalues are $\pm\sqrt{1 + x^2}$ and $\lambda_{\max}(X(x)) = \sqrt{1 + x^2}$. The minimizer is $x = 0$, with value 1. The SDP is

$$\begin{array}{ll} \text{minimize}_{x,t} & t \\ \text{subject to} & \begin{bmatrix} t - x & -1 \\ -1 & t + x \end{bmatrix} \succeq 0. \end{array}$$

This is $\sqrt{1 + x^2}$ again, the function from the DCP discussion. As a check, the 2×2 matrix is PSD exactly when $t \pm x \geq 0$ and $(t - x)(t + x) \geq 1$, that is, $t \geq \sqrt{1 + x^2}$. Writing a function as $\lambda_{\max}$ of an affine matrix is one more way to prove convexity.

Exercise 7.3. We have

$$M(w) = w_1 e_1 e_1^T + \tfrac{1}{4} w_2 e_2 e_2^T = \mathbf{diag}(w_1, w_2/4),$$

so $\lambda_{\min}(M(w)) = \min(w_1, w_2/4)$. With $w_1 + w_2 = 1$, the minimum is largest when the two entries are equal: $w_1 = w_2/4$ gives $w = (1/5, 4/5)$ and $\lambda_{\min} = 1/5$. If $w_1 > 1/5$ then $w_2/4 < 1/5$, and if $w_1 < 1/5$ then $\lambda_{\min} = w_1 < 1/5$, so this is optimal. The noisy sensor receives four times as many readings, so that both directions are known equally well.

Exercise 7.4. For feasible X, Exercise 2.5 gives $|X_{ij}| \leq 1$, so each term $w_{ij}(1 - X_{ij})/2$ is at most w_{ij}. Hence the SDP value is at most the total weight. For the 4-cycle, alternate signs around the ring: all four edges cross the cut, so the best cut equals the total weight 4. Then $4 =$ best cut $\leq$ SDP value ≤ 4, and the relaxation is tight. The same argument works for any bipartite graph.

Exercise 7.7. Since $X\mathbf{1} = 0$, we have $\|V^T\mathbf{1}\|_2^2 = \mathbf{1}^T X \mathbf{1} = 0$, so $v_1 + v_2 + v_3 = 0$. Hence the numbers $g^T v_i$ sum to zero, and (unless one of them is zero, which happens with probability zero) they cannot all have the same sign. So every rounding puts one vertex on one side and two on the other, and the cut crosses exactly two edges. The rounded cut always equals the best cut, 2. Its ratio to the SDP value is $2/(9/4) = 8/9 \approx 0.889$, consistent with the guarantee 0.878.

Exercise 8.3. If $\lambda_1 \in K_1^*$ and $\lambda_2 \in K_2^*$, then for $(s_1, s_2) \in K_1 \times K_2$, $\langle \lambda_1, s_1 \rangle + \langle \lambda_2, s_2 \rangle \geq 0$; so $K_1^* \times K_2^* \subseteq (K_1 \times K_2)^*$. Conversely, let $(\lambda_1, \lambda_2) \in (K_1 \times K_2)^*$. Every cone contains 0, so $(s_1, 0) \in K_1 \times K_2$ for every $s_1 \in K_1$, and testing against it gives $\langle \lambda_1, s_1 \rangle \geq 0$. Thus $\lambda_1 \in K_1^*$, and similarly $\lambda_2 \in K_2^*$.

Exercise 8.5. For $\min\{c^T x : Fx \le g\}$, the constraint is $g - Fx \in \mathbb{R}^m_+$, so $A = F$, $b = g$, and $K^* = \mathbb{R}^m_+$. The dual is

$$\begin{array}{ll} \text{maximize}_\lambda & -g^T\lambda \\ \text{subject to} & c + F^T\lambda = 0, \quad \lambda \ge 0. \end{array}$$

For the standard form, write $b - Ax = (g - Fx,\ x) \in \{0\}^m \times \mathbb{R}^n_+$, so $A = \begin{bmatrix} F \\ -I \end{bmatrix}$ and $b = (g, 0)$. The multiplier is $\lambda = (\nu, \mu)$ with ν free and $\mu \ge 0$. The condition $c + A^T\lambda = 0$ reads $c + F^T\nu - \mu = 0$, so $\mu = c + F^T\nu \ge 0$, and the dual objective is $-b^T\lambda = -g^T\nu$. Substituting $\nu = -\lambda'$ gives the familiar form $\max\{g^T\lambda' : F^T\lambda' \le c\}$.

Exercise 8.6. By Exercise 5.10, the constraint is $b - Ax = (x, 1) \in \mathcal{Q}^{n+1}$ with $A = \begin{bmatrix} -I \\ 0^T \end{bmatrix}$ and $b = (0, 1)$. The multiplier is $\lambda = (u, \tau) \in (\mathcal{Q}^{n+1})^* = \mathcal{Q}^{n+1}$. Then $A^T\lambda = -u$ and $b^T\lambda = \tau$, so the dual is

$$\begin{array}{ll} \text{maximize}_{u,\tau} & -\tau \\ \text{subject to} & c - u = 0, \quad \|u\|_2 \le \tau, \end{array}$$

that is, maximize $-\tau$ subject to $\tau \ge \|c\|_2$, with value $d^\star = -\|c\|_2$. For the primal, Cauchy–Schwarz gives $c^T x \ge -\|c\|_2\|x\|_2 \ge -\|c\|_2$ on the feasible set, with equality at $x = -c/\|c\|_2$. So $p^\star = d^\star = -\|c\|_2$.

Exercise 8.9. The dual is $\max\{-2\lambda : |\lambda| \le 1,\ |\lambda| \le \gamma\}$, so $\lambda^\star = -\min(1, \gamma)$ and $d^\star = 2\min(1, \gamma)$.

- $\gamma = \frac{1}{2}$: on $[0, 2]$ the primal objective is $2 - x + \frac{x}{2} = 2 - \frac{x}{2}$, which decreases; for $x > 2$ it increases, and for $x < 0$ it is larger still. So $x^\star = 2$ and $p^\star = 1 = d^\star$.
- $\gamma = 2$: on $[0, 2]$ the objective is $2 - x + 2x = 2 + x$, which increases; so $x^\star = 0$ and $p^\star = 2 = d^\star$.

For any $\gamma \ge 1$, the slope on $[0, 2]$ is $\gamma - 1 \ge 0$, so $x^\star = 0$: a large penalty sets the coefficient exactly to zero. This is the sparsity that the ℓ_1 penalty is designed to produce.

Exercise 8.13. The three blocks of $b - Ax$ are

$$x_1 + x_2 - 0.9 \in \mathbb{R}_+, \qquad (x_1,\ 0.2x_2,\ 1 - x_1 - x_2) \in \mathcal{Q}^3, \qquad (x_1, x_2) \in \mathbb{R}^2_+,$$

so $b = (-0.9;\ 0, 0, 1;\ 0, 0)$ and the rows of A are $(-1, -1)$; $(-1, 0)$, $(0, -0.2)$, $(1, 1)$; and $(-1, 0)$, $(0, -1)$. Dual feasibility: $K^* = \mathbb{R}_+ \times \mathcal{Q}^3 \times \mathbb{R}^2_+$ by Proposition 8.2 and Exercise 8.3. The first and last blocks of λ are nonnegative, and $\|(-0.15, -0.75)\|_2 \approx 0.765 \le 0.85$, so the middle block lies in $\mathcal{Q}^3$. Next,

$$A^T\lambda = 1\begin{pmatrix} -1 \\ -1 \end{pmatrix} - 0.15\begin{pmatrix} -1 \\ 0 \end{pmatrix} - 0.75\begin{pmatrix} 0 \\ -0.2 \end{pmatrix} + 0.85\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 + 0.15 + 0.85 \\ -1 + 0.15 + 0.85 \end{pmatrix} = 0,$$

and $b^T\lambda = -0.9 + 0.85 = -0.05 < 0$. By Proposition 8.11, the problem is infeasible.

Exercise 8.14.

(a) The problem is $\max\{v^T u : (u, \rho) \in \mathcal{Q}^{k+1}\}$.

(b) Attach $(z, \tau) \in (\mathcal{Q}^{k+1})^* = \mathcal{Q}^{k+1}$. For feasible u,

$$\mathcal{L}(u, z, \tau) = v^T u + z^T u + \tau\rho \geq v^T u,$$

since $\langle (z, \tau), (u, \rho) \rangle \geq 0$. The supremum over u is finite only if $z = -v$, and then it equals $\rho\tau$. The dual is $\min\{\rho\tau : \|v\|_2 \leq \tau\}$, with value $\rho\|v\|_2$, which agrees with (1).

(c) Apply this with $v = P^T x$. If some τ satisfies $\|P^T x\|_2 \leq \tau$ and $\bar{a}^T x + \rho\tau \leq b$, then by weak duality the worst case is at most $\rho\tau$, so the robust constraint holds. Conversely, if the robust constraint holds, take $\tau = \|P^T x\|_2$; this is the dual optimum, and strong duality says it equals the worst case.

(d) The "if" direction needs only weak duality. The "only if" direction uses strong duality with an attained dual optimum. Slater's condition holds because $u = 0$ is strictly feasible: $\|0\|_2 < \rho$ when $\rho > 0$, so $(0, \rho) \in \mathbf{int}\, \mathcal{Q}^{k+1}$, and the primal value is finite because the ball is bounded.

10 Key concepts

- *Support function* $\sigma_{\mathcal{U}}(v) = \sup_{u \in \mathcal{U}} u^T v$; for a Euclidean ball of radius ρ it is $\rho\|v\|_2$, and for a box it is $\rho\|v\|_1$.
- *PSD matrices*: $X \succeq 0$ iff $z^T X z \geq 0$ for all z iff all eigenvalues are nonnegative iff $X = VV^T$; Rayleigh quotient characterization of $\lambda_{\max}$ and $\lambda_{\min}$.
- *Robust constraint*: one x feasible for every error in an uncertainty set; its *exact robust counterpart* for ellipsoidal uncertainty is $\bar{a}^T x + \rho\|P^T x\|_2 \leq b$, an SOC constraint.
- *Chance constraint* with Gaussian data: an SOC constraint with $\rho = \Phi^{-1}(\eta)$ when $\eta \geq \frac{1}{2}$.
- *Exact reformulation, restriction, relaxation*: same value; smaller feasible set (feasible but conservative); larger feasible set (a bound). Always ask: relative to which model?
- *Convex cone*; the standard cones $\{0\}$, $\mathbb{R}^m_+$, $\mathcal{Q}^{m+1}$, $\mathbf{S}^m_+$, and $K_{\exp}$; *product cones.*
- *Conic program* $\min\{c^T x : b - Ax \in K\}$; the hierarchy LP $\subset$ QP $\subset$ SOCP $\subset$ SDP.
- *Epigraph trick* and *conic representations*; lifting with auxiliary variables; why upper-bound variables cannot enforce lower bounds.
- *Derivative-oracle versus conic solvers*: what each needs, what each guarantees, and when to use each.
- *Disciplined convex programming*: composition rules are sufficient, not necessary; canonicalization by a modeling language.
- *Equivalent formulations can cost very different amounts*: preserve structure (SOC rather than PSD, native QP rather than SOCP).

- *Linear matrix inequalities*: $\lambda_{\max}(X) \le t \iff tI - X \succeq 0$; minimizing $\lambda_{\max}$ of an affine matrix; E-optimal experiment design.
- *MaxCut and the Shor relaxation*: lift $X = xx^T$, drop $\mathrm{Rank}(X) = 1$, obtain an upper bound; Goemans–Williamson randomized rounding and the 0.878 guarantee.
- *Dual cone* K^*; self-duality of $\mathbb{R}^m_+$, $\mathcal{Q}$, and $\mathbf{S}_+$; free multipliers for equalities.
- *Conic dual* $\max\{-b^T\lambda : c + A^T\lambda = 0,\ \lambda \in K^*\}$; weak duality; Slater's condition; complementary slackness $\langle \lambda, s \rangle = 0$.
- *Dual norms*: $\ell_2 \leftrightarrow \ell_2$, $\ell_1 \leftrightarrow \ell_\infty$.
- *Certificates*: the duality gap certifies near-optimality; a conic Farkas certificate proves infeasibility.

11 For the quiz

What you should be able to do

1. Derive the robust counterpart of a linear constraint under ball or box uncertainty, by computing the worst-case error with Cauchy–Schwarz or Hölder's inequality.
2. Write a robust or chance constraint as membership of an affine expression in a second-order cone, and identify when a chance constraint is convex ($\eta \ge \frac{1}{2}$).
3. Classify a new problem as an exact reformulation, a restriction, or a relaxation of a given one, and say which direction of bound a relaxation gives.
4. Put a problem with norms, absolute values, or maxima into conic form: introduce epigraph variables, list every constraint with its cone, and argue exactness in both directions.
5. Check whether an expression is DCP, and rewrite a convex non-DCP expression in DCP form.
6. Translate eigenvalue bounds into linear matrix inequalities, and recognize $\lambda_{\max}$ minimization and E-optimal design as SDPs.
7. Write the MaxCut lifting and SDP relaxation, and explain which constraint is dropped and why the result is an upper bound.
8. Compute simple dual cones and prove that $\mathcal{Q}$ is self-dual.
9. Derive the dual of a conic program by the three-step recipe: attach multipliers in the dual cone, check the bound, and optimize the Lagrangian over the primal variables.
10. Verify a certificate by hand: a dual feasible point and its bound, a zero gap, or a Farkas certificate of infeasibility.

Common mistakes

- Forgetting $t \geq 0$ when squaring a norm constraint $\|z\|_2 \leq t$.
- Using an auxiliary upper-bound variable to enforce a lower bound, as in $\|x\|_1 \geq 1$.
- Calling the MaxCut SDP a reformulation: it is a relaxation, and its value is an *upper* bound for a maximization problem.
- Treating a merely feasible SDP matrix as a bound; only the optimal value, or a dual feasible point, bounds the best cut.
- Giving a cone multiplier the wrong sign: it must lie in K^*, and equality multipliers are free.
- Forgetting the condition $c + A^T\lambda = 0$; without it the dual function is $-\infty$.
- Concluding that a problem is nonconvex because a DCP check failed.
- Reading a small duality gap as a certificate without first checking primal and dual feasibility.
- Assuming $\rho = 2$ gives a 97.7% guarantee without a distributional model.

Self-test

Answer these without notes; brief answers follow.

1. What is preserved when we replace $\|Ax - b\|_2$ by an epigraph variable t with $\|Ax - b\|_2 \leq t$?
2. Why does ellipsoidal uncertainty lead to a second-order cone constraint?
3. Which step in the MaxCut formulation changes the feasible decisions?
4. Why must a conic multiplier lie in the dual cone?
5. Is $\{(x, t) : \|x\|_1 \leq t\}$ representable using only nonnegative orthants? Is $\{X : \lambda_{\max}(X) \leq 1\}$?
6. The robust plan is infeasible for the nominal model. True or false?
7. What is the dual of $\max\{\langle C, X\rangle : \mathbf{tr}\, X = 1,\ X \succeq 0\}$, and what is its optimal value?

Answers. (1) The optimal value and the optimal x: every x lifts to $(x, \|Ax-b\|_2)$, and every lifted point is no better than its x. (2) The worst-case error over a Euclidean ball is the support function $\rho\|P^Tx\|_2$, a norm of an affine expression. (3) Dropping $\mathrm{Rank}(X) = 1$; the lifting $X = xx^T$ with the rank constraint is exact. (4) So that each added Lagrangian term is nonpositive on feasible points, which keeps the Lagrangian a lower bound (for a minimization). (5) Yes: $-s \leq x \leq s$, $\mathbf{1}^Ts \leq t$. Yes, but with a PSD cone, not an orthant: $I - X \succeq 0$. (6) False. The robust constraint implies the nominal one (take $u = 0$), so the robust model is a restriction of the nominal model. (7) Build an upper bound with a free multiplier μ for $\mathbf{tr}\, X = 1$ and $Z \succeq 0$: $\mathcal{L} = \langle C, X\rangle + \mu(1 - \mathbf{tr}\, X) + \langle Z, X\rangle$, bounded in X only if $C - \mu I + Z = 0$. The dual is $\min\{\mu : \mu I - C \succeq 0\}$, whose value is $\lambda_{\max}(C)$. (The primal attains it at $X = qq^T$ for a top eigenvector q.)

Further reading. Boyd and Vandenberghe, *Convex Optimization*, Sections 4.3–4.6 (conic form and generalized inequalities), 5.9, and 7.5 (experiment design); the CVXPY tutorial on disciplined convex programming; Lobo, Vandenberghe, Boyd, and Lebret, "Applications of second-order cone programming"; Ben-Tal, El Ghaoui, and Nemirovski, *Robust Optimization* (2009); Goemans and Williamson, "Improved approximation algorithms for maximum cut and satisfiability problems using semidefinite programming," *JACM* (1995). The scripts in `conic-v1-assets/` reproduce the numbers and figures in the examples.